



Getting Started with NetApp Ransomware Resilience

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1 Introduction

Costly ransomware attacks are routinely featured in the news and companies know that it's critical to protect their valuable data from this threat. While many businesses have responded by adopting ransomware protection tools, such as those offered by NetApp ONTAP, it can be difficult to manage protection and recovery across many storage clusters. This is especially true when these clusters serve different functions and are in different locations.

Companies need a solution that makes it easy to ensure all of their workloads and disparate data are protected and that, should an attack occur, the malicious activity will be highly visible and easy to respond to. NetApp Ransomware Resilience aims to solve this challenge for companies that utilize ONTAP-based storage, whether in their datacenter or in the cloud.

NetApp Ransomware Resilience is designed to map all file share and application workloads and make it easy to achieve your infrastructure-wide protection goals. NetApp Ransomware Resilience is a policy-based solution that simplifies adding layered defenses, combining many of the ransomware protection features that NetApp offers into a bundle of policies called a protect “strategy”. Once your data is protected, NetApp Ransomware Resilience centralizes alerts of potential ransomware activity on your ONTAP storage, and provides fast recovery options.

 **Note:** NetApp now offers the NetApp Ransomware Recovery Guarantee. This guarantees that you will be able to recover your protected NetApp Snapshot copies on eligible arrays, and if you can't, your organization will be compensated. To receive that guarantee, you must purchase the NetApp Ransomware Recovery Assurance Service at the time you purchase a qualified storage array. In addition, AutoSupport must be enabled on each qualified storage array. NetApp experts will ensure that the mandatory protection and recovery features are properly configured and assist with recovery of the data in the event of an attack.

For more information on the NetApp Ransomware Recovery Guarantee, check the [References](#) section.

In this lab, you take steps as a storage administrator to protect your company against ransomware using NetApp Ransomware Resilience. After you map out your workloads, you will protect them by assigning policies that enable features such as Autonomous Ransomware Protection (ARP), scheduled snapshot copies, and FPolicy to block known ransomware file extensions and monitor for anomalous user behavior. You will then use the tools provided by NetApp Ransomware Resilience to respond to either a simulated ransomware attack or a data breach.

2 Lab Activities

In this lab you explore the following topics:

- [Prepare Your Lab Environment](#)
- [Protect Your Workloads](#)
 - [Discover Workloads](#)
 - [Add a Backup Destination](#)
 - [Activate User Activity Agent](#)
 - [Enable Protection](#)
- [Option 1: Respond to a Ransomware Attack](#)
 - [Detect a Ransomware Attack](#)
 - [Recover from a Ransomware Attack](#)
- [Option 2: Respond to a Data Breach](#)

2.1 Prepare Your Lab Environment

In this lab, you use NetApp Console as the control plane to centrally manage the hybrid cloud. You will log in to the Console with a user account that has administrative privileges for your “organization”, which is the top-level organizational unit in NetApp Console, and which might for example correspond to a business. Within an organization, administrators manage storage resources in isolated workspaces called “projects”. For example, a group of users might deploy and manage Cloud Volumes ONTAP systems in a project that is not visible to other users.

The Console uses a web interface that lets you navigate between your organizations and projects, as well as NetApp Console “agents” associated with each project. Console agents are servers that provide connectivity between your on-premises or cloud environments with the NetApp Console platform. All of these components are part of the Identity and Access Management (IAM) system used by the Console.

In this activity, you navigate to the in-lab account information where you can then use those supplied credentials to log into the Console. This lab has a dedicated NetApp Console organization, project, and agent that were created before you connect to the lab. The account(s) for use with NetApp Console IAM have been pre-created in order to facilitate quicker access to the lab environment.



Important: You must perform the steps in this section to proceed with subsequent activities.

1. After connecting to the Jumphost, look for the Automation shortcut. If it appears as a blue icon with two arrows, the lab automation that configures your lab is still running. You can double-click the icon to check the status.

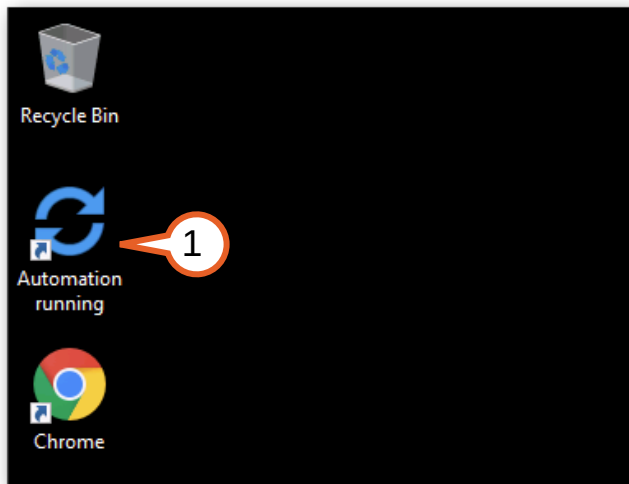


Figure 2-1:

2. When the automation completes the shortcut changes to a green check.
Double-click the **Automation completed** shortcut to launch the Lab Control Panel in the Chrome browser.



Figure 2-2:

3. You are on the Lab Status page. Click **Accounts**.

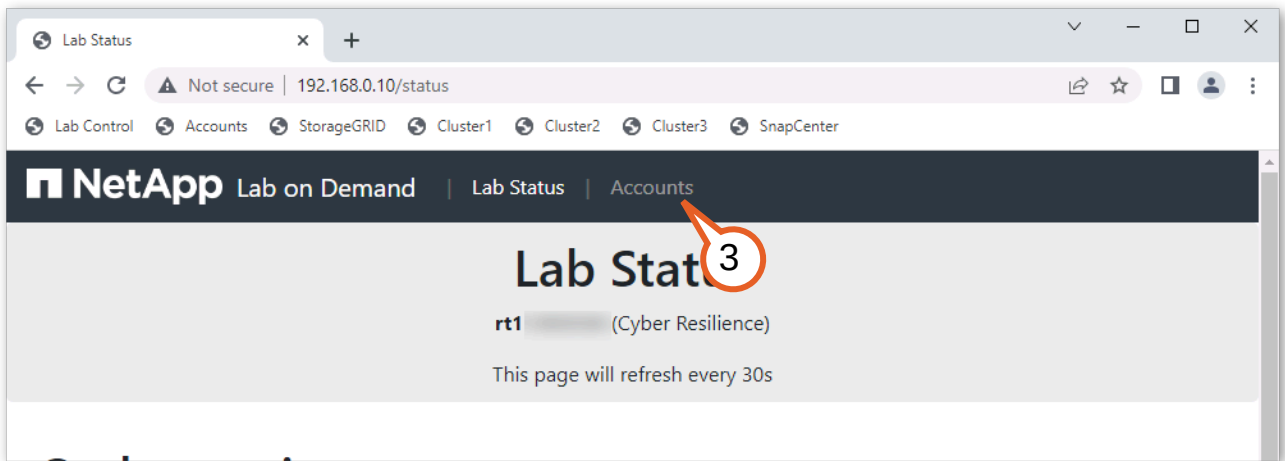


Figure 2-3:

4. Click the link to the right of the Environment Information row (the box with arrow pointing up) to go to the console.

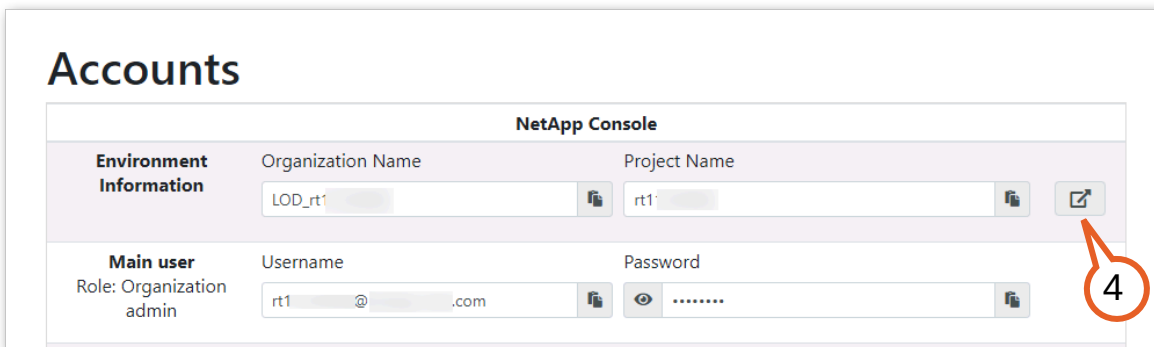


Figure 2-4:

5. In a new tab, you are prompted to log in to the Console.
Copy, then paste the Main user Username from the Accounts page to the Email field on the NetApp Console login page.

6. Click **Next**.

Accounts

NetApp Console

Environment Information

Organization Name: LOD_rt1

Project Name: rt

Main user

Role: Organization admin

Username: rt11@...com

Password:

Additional user

Username: ...

Password: ...

Log in to NetApp Console

Use your Console or NetApp Support Site (NSS) credentials

rt11@...com

Next

Don't have a Console or NSS Account? [Sign up](#)

Figure 2-5:

7. Now copy, then paste the Main user Password to the password field.

8. Click **Log in**.

Accounts

NetApp Console

Environment Information	Organization Name	Project Name
	LOD_rt	rt1

Main user Role: Organization admin	Username	Password
	rt_@.com	

Log in to NetApp Console

← rt_@netapp-test.com

.....

Log in

[Forgot password?](#)

Figure 2-6:

9. Upon signing in you are greeted with a dashboard providing an overview of your environment. Click the **menu icon** in the top-left.

10. From the menu, select **Administration > Identity and Access**.

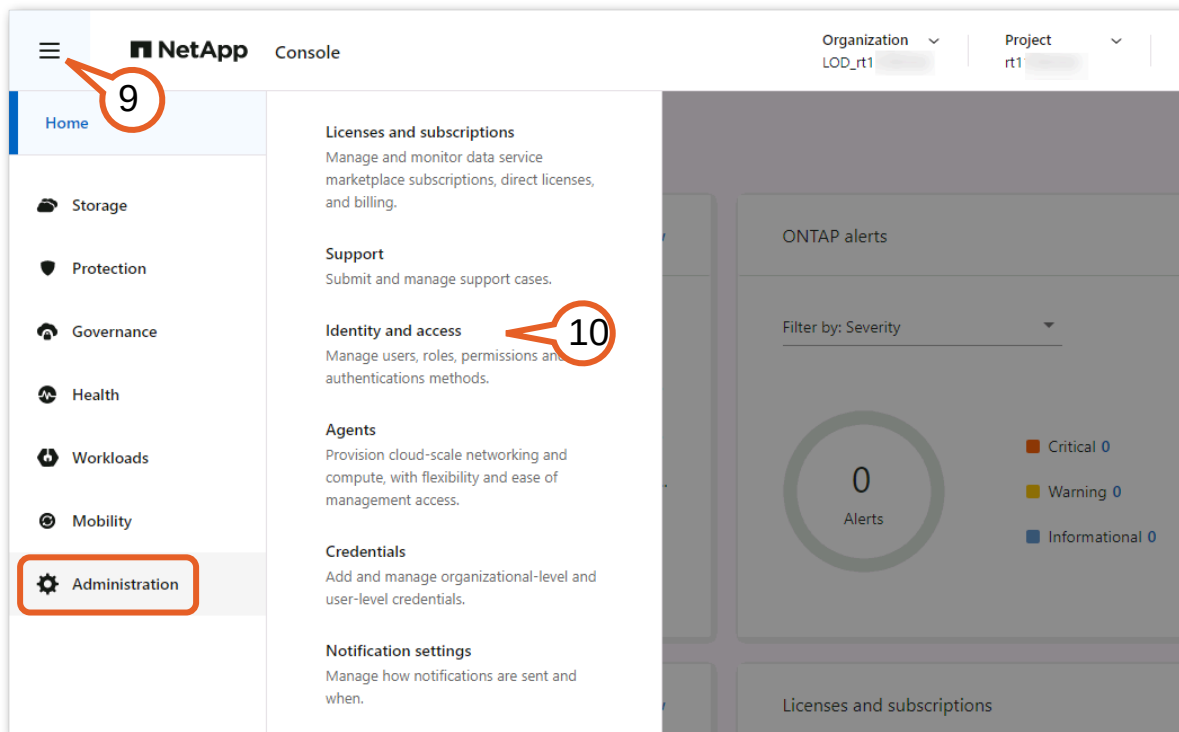


Figure 2-7:

11. In order to perform the actions in this lab, you will need to give your NetApp Console administrative account appropriate permissions for the Ransomware Resilience service. As a user with the Organization admin role, you are able to give yourself the necessary roles.

Click **Members** from the menu on the left.

12. Locate your user (rt1*****), and click the **menu icon** to the right.

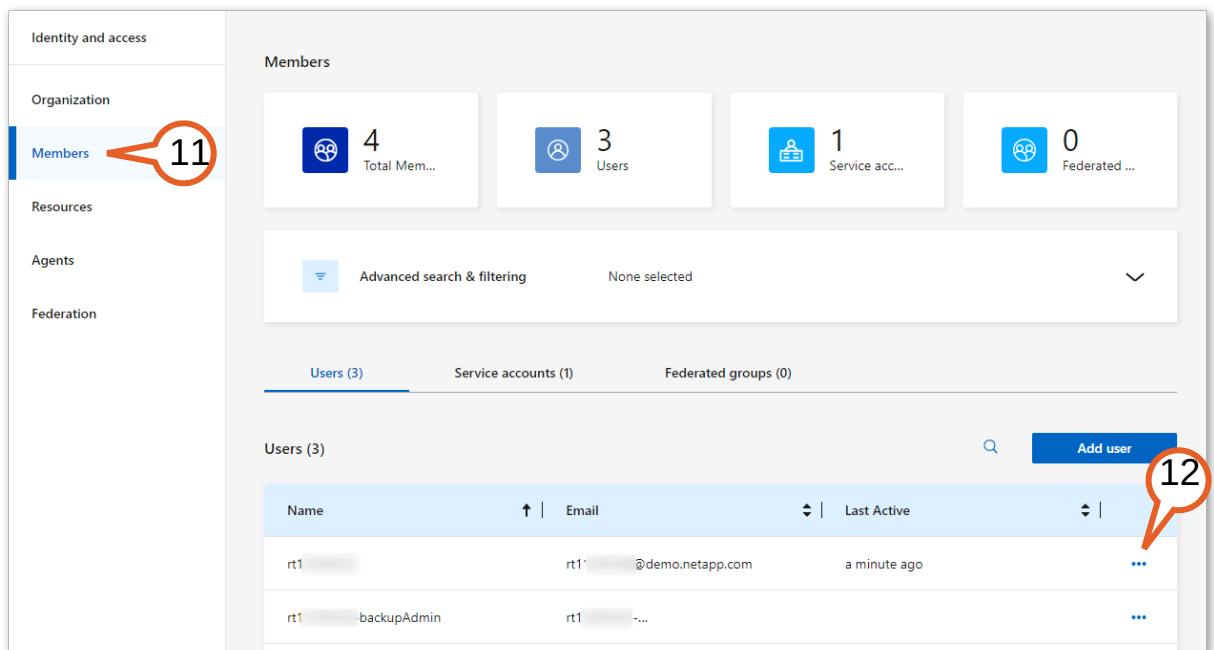


Figure 2-8:

13. From the menu that appears, click **Add a role**.

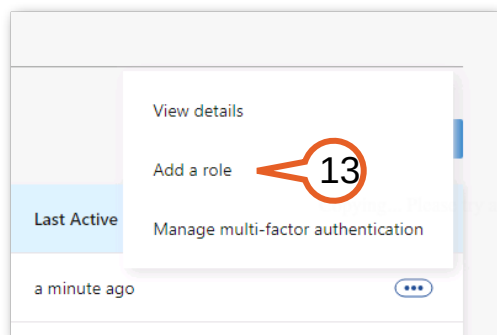


Figure 2-9:

14. Under Organization, folder or project, select both your organization and the project created for the lab.

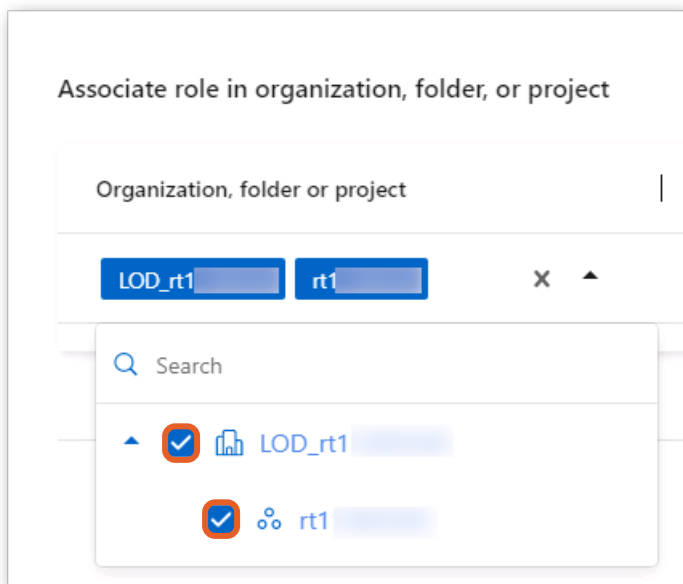


Figure 2-10:

15. Under Category, select **Data Service**.
16. Under Role, select **Ransomware resilience user behavior admin**. You will need to scroll down the list.



Note: The other needed role, Ransomware resilience admin, is already added to your account.

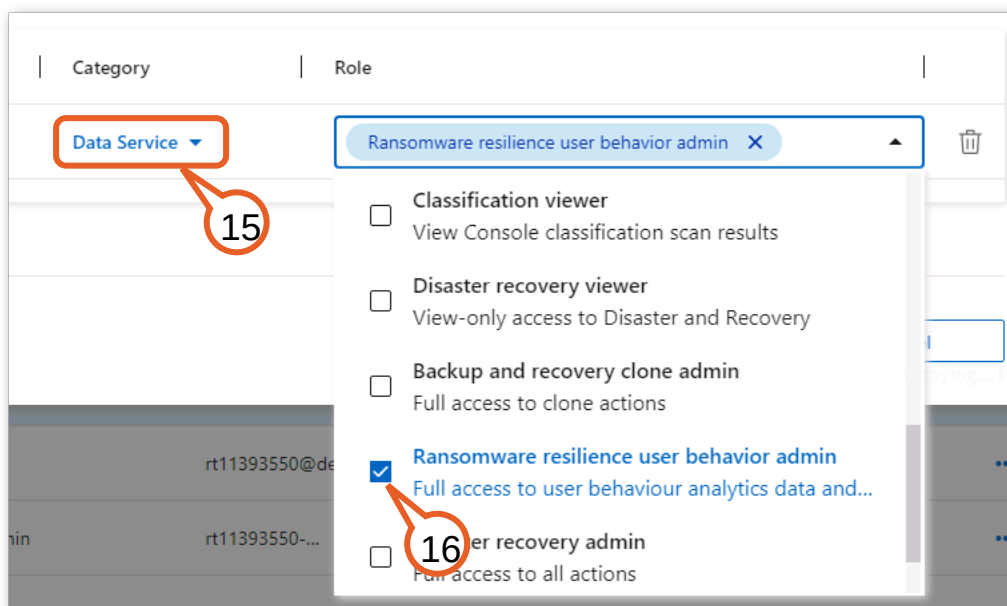


Figure 2-11:



Important: Ensure that **Ransomware resilience user behavior admin** is selected, and not the similar “Ransomware resilience user behavior viewer”. If the “admin” role is not added you will be prevented from completing activities later in this lab..

17. Click outside the drop-down menu to close it and then click **Add new roles**.

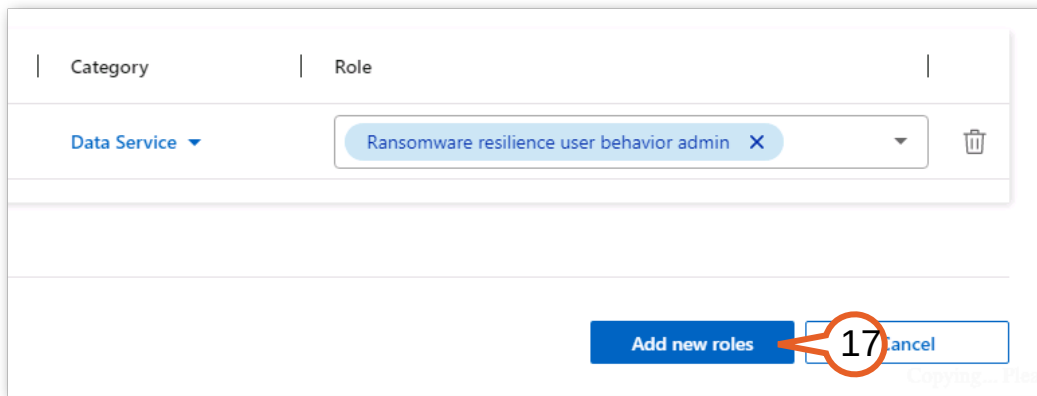


Figure 2-12:

18. Click the menu icon for your user (rt1*****), and select **View details** from the menu.

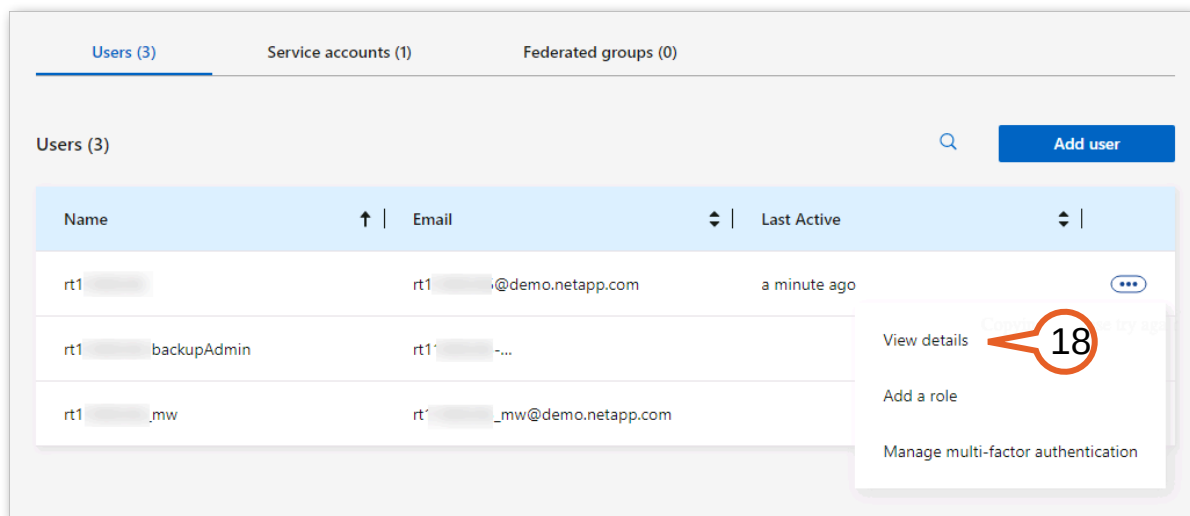


Figure 2-13:

19. Click the **arrow icon** next to the “LOD_rt1*****” organization to expand it.

20. Confirm that both **Ransomware resilience admin** and **Ransomware resilience user behavior admin** are selected.

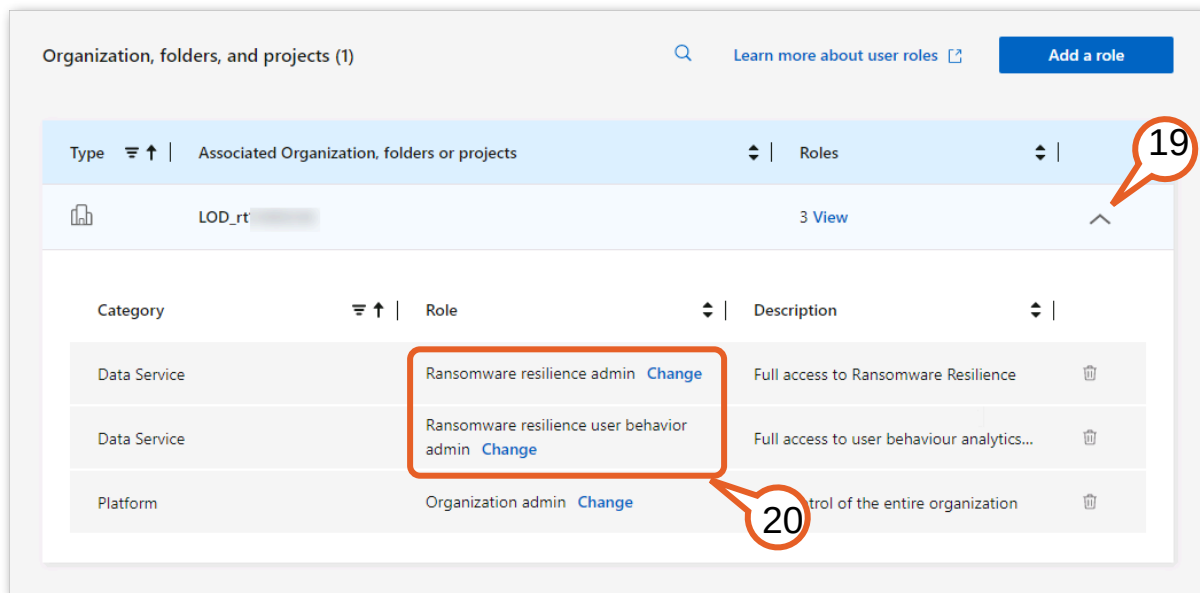


Figure 2-14:

You are now signed into the Console administrative account for this lab and the account has the appropriate roles needed for the lab activities. In the next activity, you will begin to explore NetApp Ransomware Resilience.

2.2 Protect Your Workloads

Companies concerned about ransomware attacks should first seek to protect their workload data. This allows workloads to be recovered in case an attack occurs or even prevent damage from occurring in the first place. Storage administrators need to enable attack detection and proactively create flexible recovery points for data using snapshot copies and backups. In order to achieve their protection goals throughout their storage infrastructure, they need confidence that no workload is overlooked. NetApp Ransomware Resilience makes this easy with automated workload discovery, integrated backup management, and integrations with powerful NetApp ransomware protection features (including Autonomous Ransomware Protection and User behavior analytics which will be explored later in this lab).

In this section of the lab, you discover workloads in your NetApp Console Project, add a StorageGRID backup destination, set up a User behavior analytics agent, and then protect your workloads using a strategy that combines all of these protections. This section has the following activities:

Lab Activities
Discover Workloads
Add a Backup Destination
Activate User Activity Agent
Enable Protection

2.2.1 Discover Workloads

Once enabled, the NetApp Ransomware Resilience data service begins to discover the workloads in your environment. During discovery, NetApp Ransomware Resilience analyzes all volumes and files in working environments across all Console Agents and projects within an organization.

NetApp Ransomware Resilience assesses MySQL applications, Oracle applications, VMware datastores, and file shares.



Important: Workloads with volumes that use FlexGroup or iSCSI won't be discovered. Support for additional workloads is planned for the future. For more information about what workloads are supported, check the [References](#) section.

The service assesses the existing protection level including the current backup protection, Snapshot copies, and NetApp Autonomous Ransomware Protection (ARP) options. Based on the assessment, the service then recommends how to improve your ransomware protection.

In this activity, you initiate automated workload discovery and then examine the results, including how NetApp Ransomware Resilience detects and prioritizes your workloads.

1. Click the **menu button** in the top-left.
2. In the menu that appears, navigate to **Protection > Ransomware Resilience**.

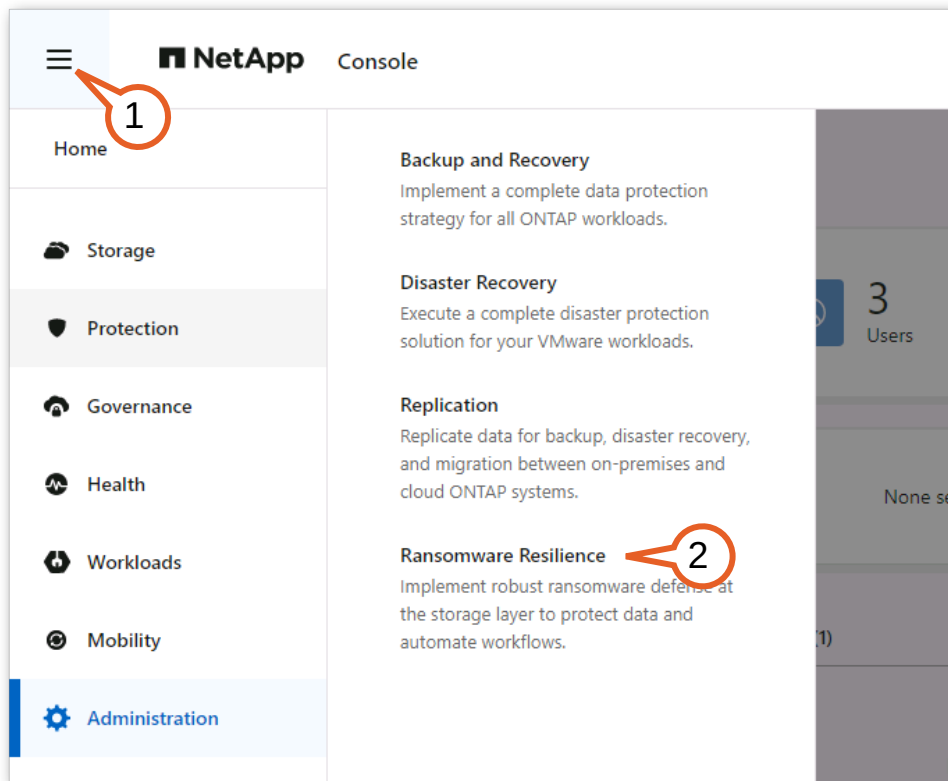


Figure 2-15:

3. You are taken to the landing page for NetApp Ransomware Resilience. Click **Start 30-day free trial**.



Note: NetApp Ransomware Resilience offers multiple licensing options.

You can use the following license types:

- 30-day free trial (which you will use for this lab).
- Purchase a pay-as-you-go (PAYGO) subscription with Amazon Web Services (AWS) Marketplace.
- Bring your own license (BYOL), which is a NetApp License File (NLF) that you obtain from your NetApp Sales Rep.

Ransomware Resilience requires a NetApp ONTAP One license.

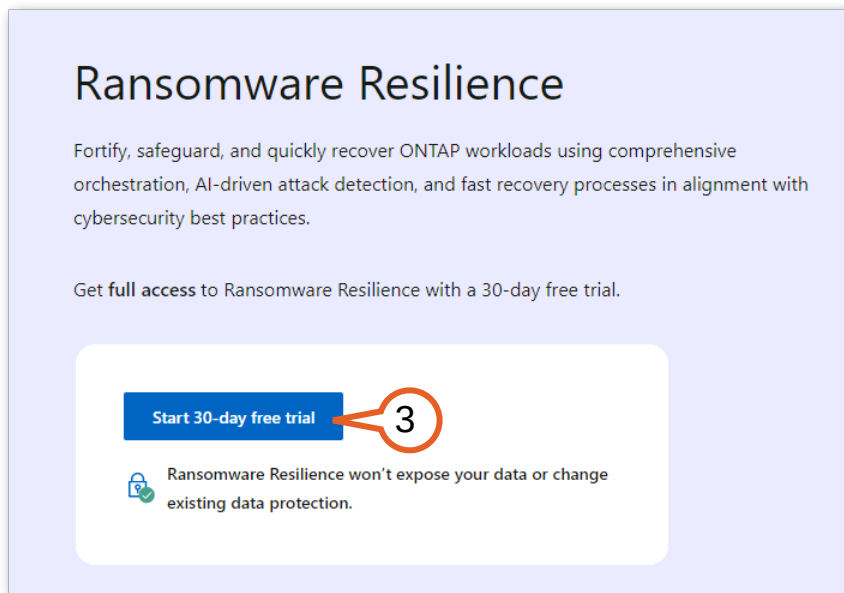


Figure 2-16:

4. The Discover Workloads view is shown and it lists the Console Agents. Click **Select systems** to the right of onpremAgent.

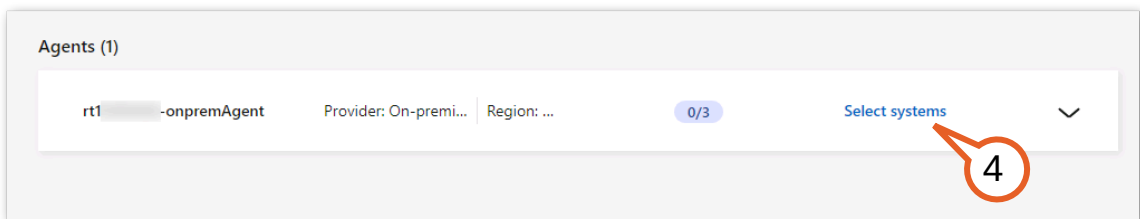


Figure 2-17:

5. Click the **check boxes** to select all three systems.

6. Click **Discover**.

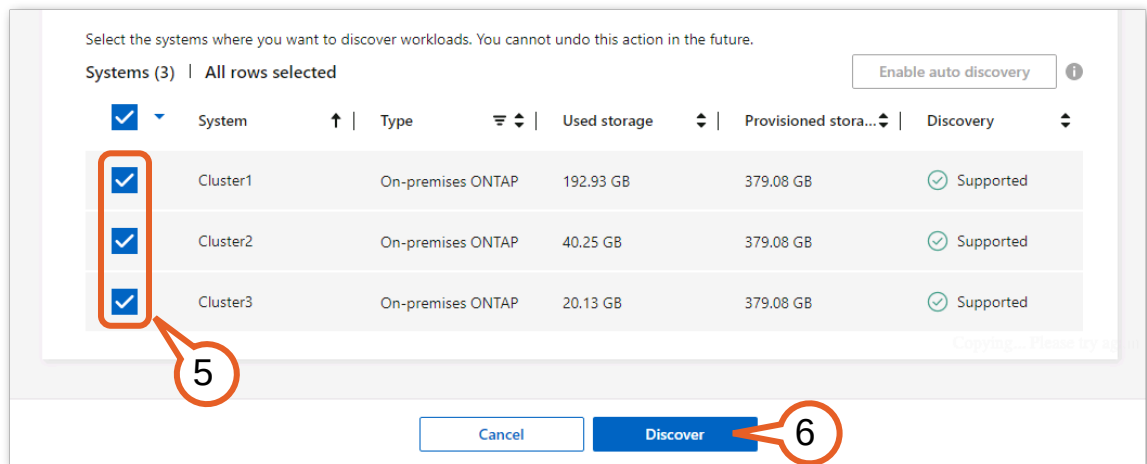


Figure 2-18:

7. When the discovery process completes, the results show that several workloads were found.

8. Click **Go to Dashboard**.

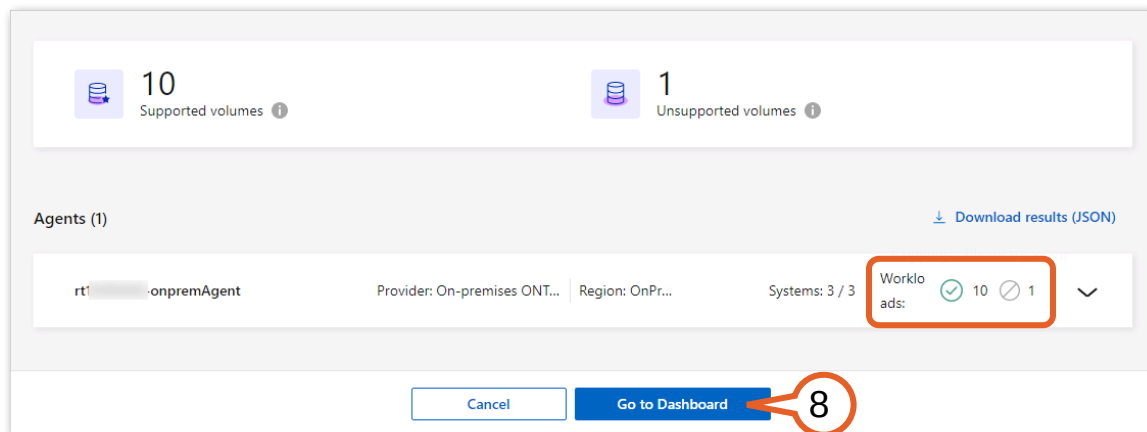


Figure 2-19:

9. You should see the dashboard view of NetApp Ransomware Resilience. This dashboard displays the results of the workload discovery process.

You can dismiss the notification about the length of the trial by clicking the **close button**.

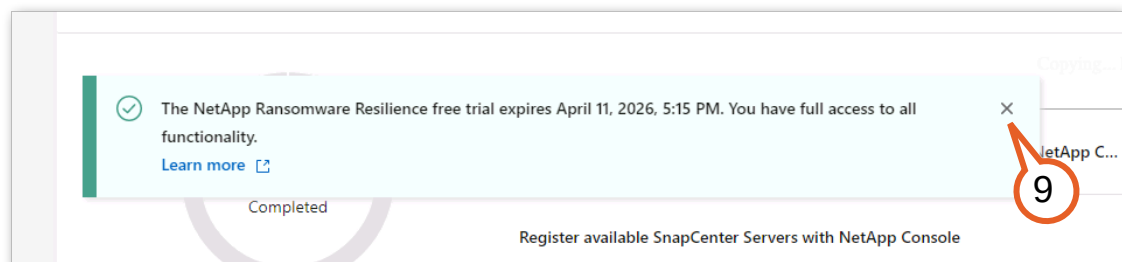


Figure 2-20:

10. The Recommended actions panel displays a list of to-do items corresponding to unprotected workloads and additional NetApp Ransomware Resilience configuration steps. The to-do list provides you with a link to review and fix each issue. You will address many of these items throughout the lab.
11. Click the **Protection** tab.

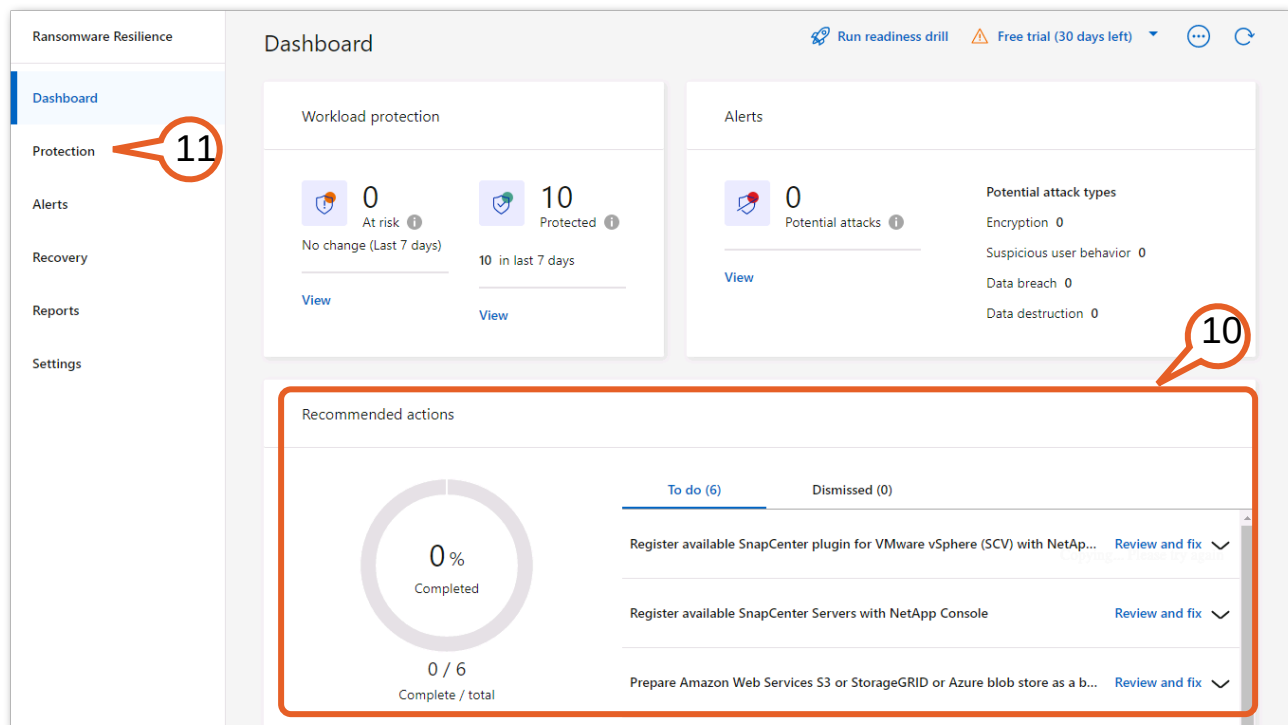


Figure 2-21:

12. Notice that the discovered workloads are displayed in a table.

In order to take full advantage of all the features that Ransomware Resilience provides, you need to perform some additional setup tasks before we start protecting workloads.

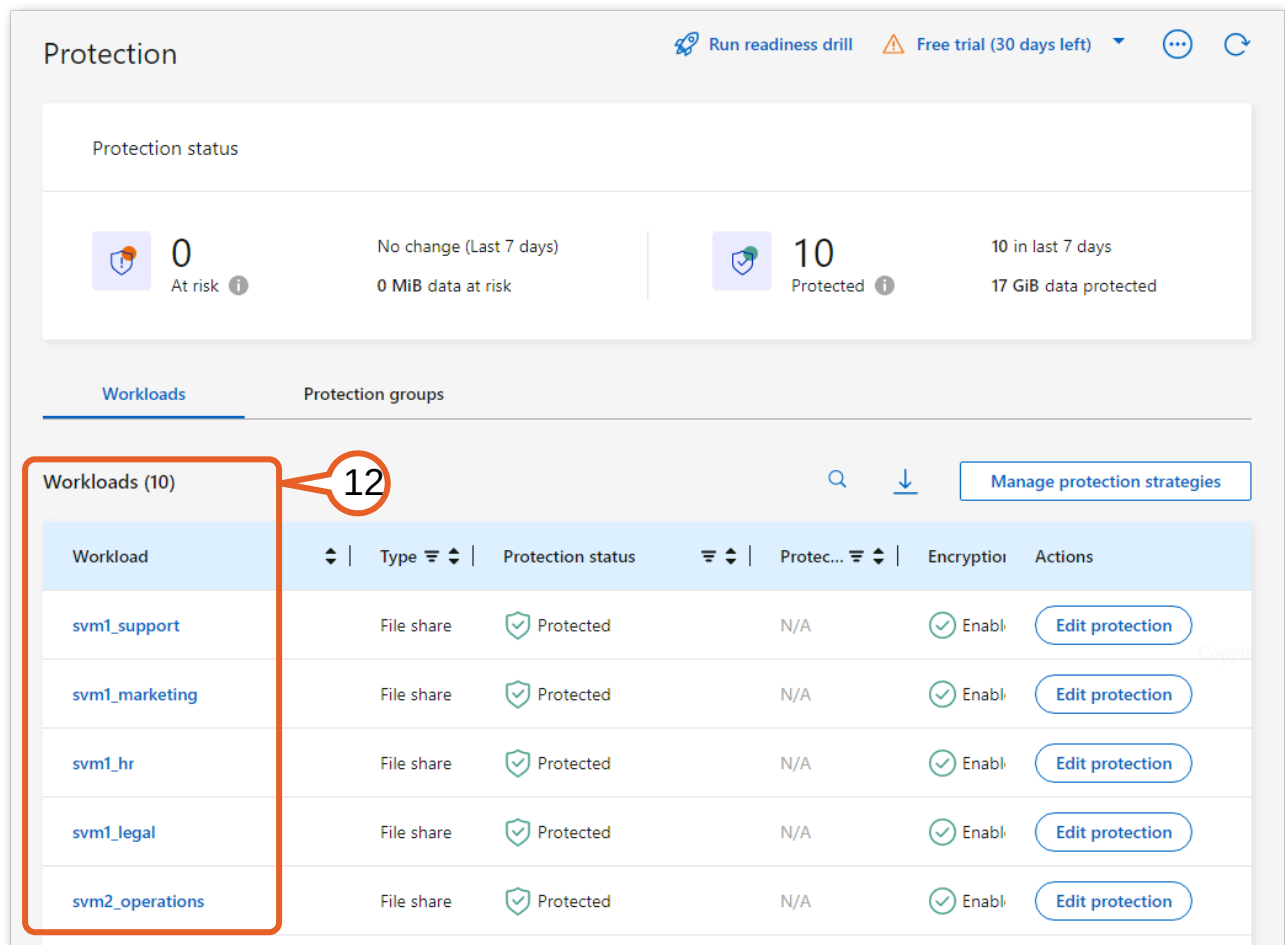


Figure 2-22:

With your workloads discovered, in the next activity you will add an object store to use as a backup destination within NetApp Ransomware Resilience.

2.2.2 Add a Backup Destination

An important part of a NetApp Ransomware Resilience protection strategy is backing up your workload data. Before you can begin to protect your workloads, you need to configure a backup destination. For file share workloads, this means backing up to an object store in AWS, Azure, or StorageGRID. In this activity, you add a StorageGRID object store that is running in your lab environment as a backup destination.

1. Select **Settings** from the menu.

2. On the settings page, click the **View** button under Backup destinations.

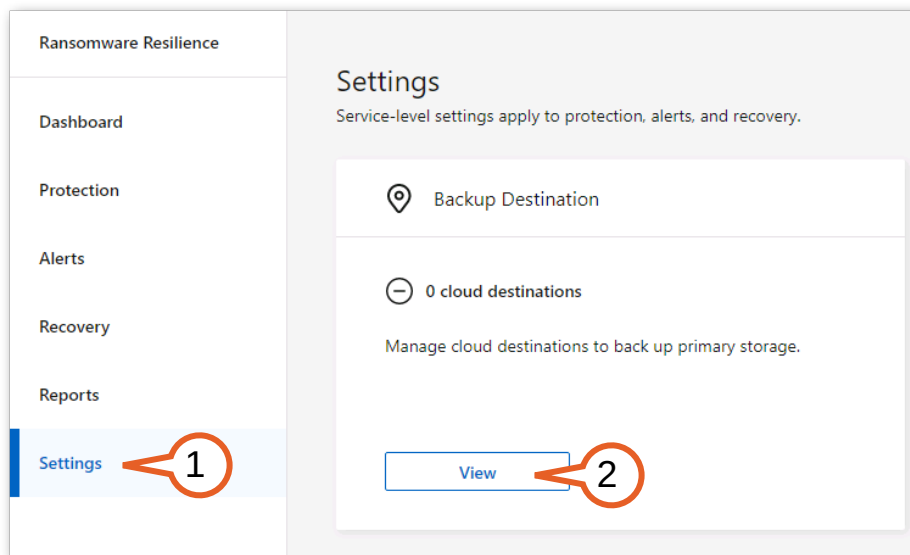


Figure 2-23:

3. You are taken to the Backup destinations page. Click **Add** to begin adding your backup destination.

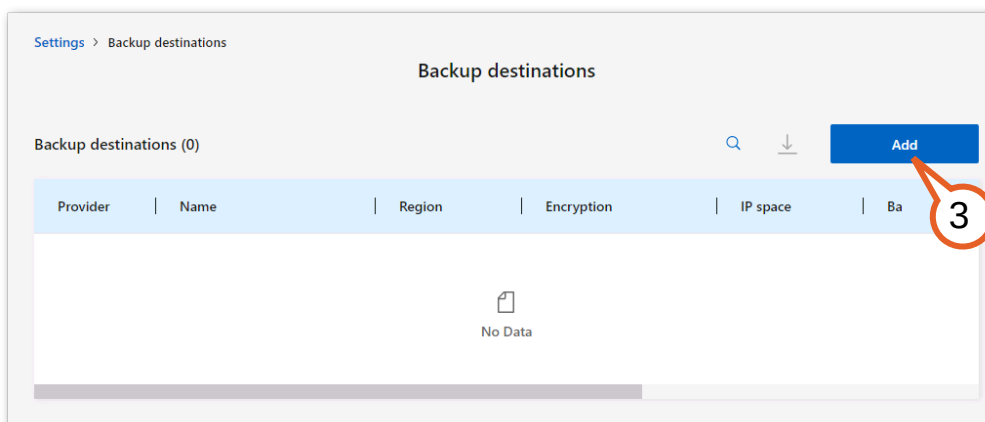


Figure 2-24:

4. In the Name field, enter **sg1**.

5. Click **Provider** to select the provider.

Add backup destination

Name

Name of backup destination

sgt

Provider

Action required

Figure 2-25:

6. Backups are saved to object storage in the cloud. This also includes on-premises StorageGRID system as an option to store backups. Click **StorageGRID**.
7. Click **Provider settings**.

Provider

Select a provider to back up to the cloud.

aws

Amazon Web Services

Microsoft Azure

Google Cloud Platform

StorageGRID

Provider settings

Action required

Figure 2-26:

8. Under provider settings, enter these values:
 - Gateway node fully qualified domain name (FQDN): **dc1-adm1.demo.netapp.com**
 - Port: **10443**

9. Click the **Accounts** tab which should still be open in your in-lab Chrome browser.

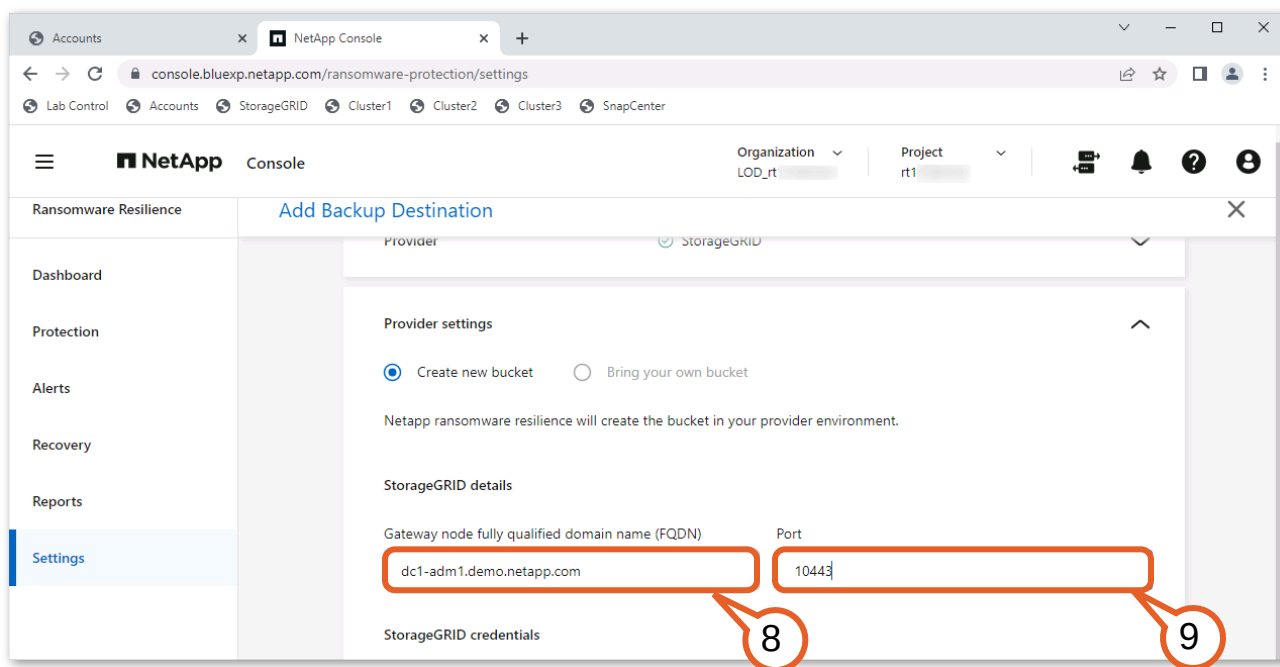


Figure 2-27:

10. Scroll down to the StorageGRID API Keys, and click the **copy buttons** for both the Secret Key and the Access Key, pasting them in the appropriate fields in the NetApp Console tab.

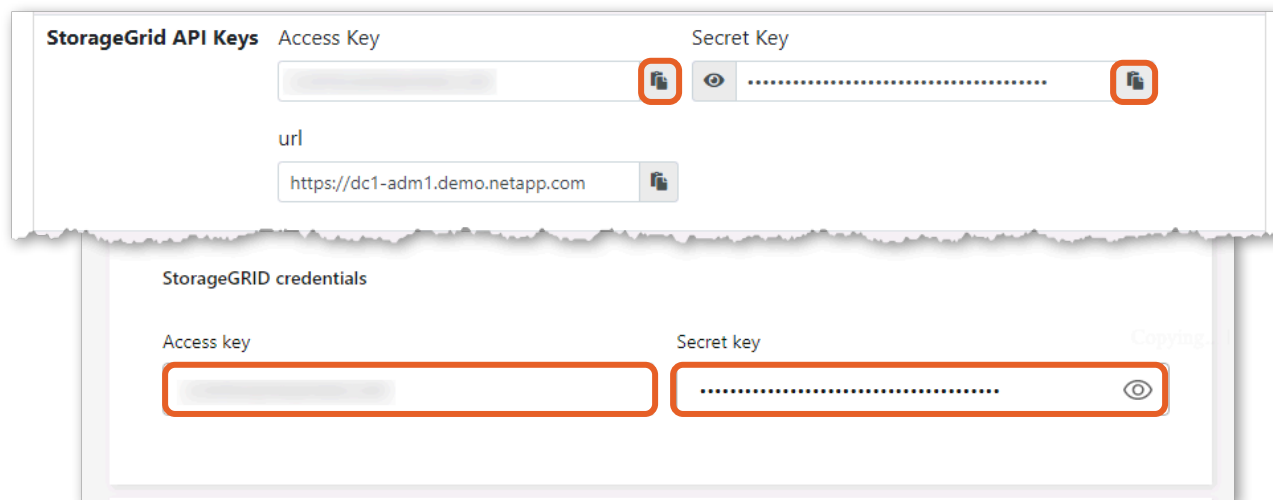


Figure 2-28:

11. Now click **Add** at the bottom of the backup destination wizard.



Note: Notice that NetApp Ransomware Resilience allows you to enable NetApp Data lock which prevents backup deletion and performs ransomware scans of the backups. For more information about Data lock, check the [References](#).

Networking IPspace: default

Backup lock None

Cancel Add 11

Figure 2-29:

12. You should now see your StorageGRID cluster in the backup destination table, as well as a notification stating the destination was added successfully.

Click the **close icon** for the notification to dismiss it.

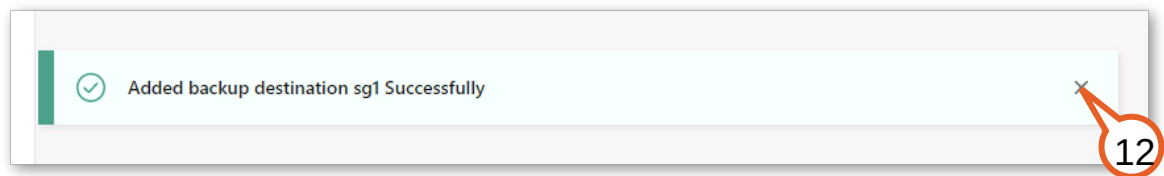


Figure 2-30:

With your backup destination added, in the next activity you will set up User Activity monitoring.

2.2.3 Activate User Activity Agent

With user activity detection, NetApp Ransomware Resilience enables you to address ransomware events at the user-level, and is designed to monitor for and allow you to take action on data breaches or large-scale deletion.

NetApp Ransomware Resilience uses AI-driven data breach detection by monitoring for suspicious user activity. Sharp increases in read activity and read activity access patterns are used to determine malicious intent. You will have an opportunity to use NetApp Ransomware Resilience to respond to a data breach later in this lab.

Ransomware Resilience detects suspicious user activity by analyzing user activity events generated by FPolicy in ONTAP. To collect user activity data, you need to deploy one or more user activity agents. The user activity agent is a Linux server or VM with connectivity to devices on your tenant.

In this activity, you set up a User activity agent using a Linux VM within your lab environment.

1. Click **Settings**.

2. Under User activity, click **Activate**.

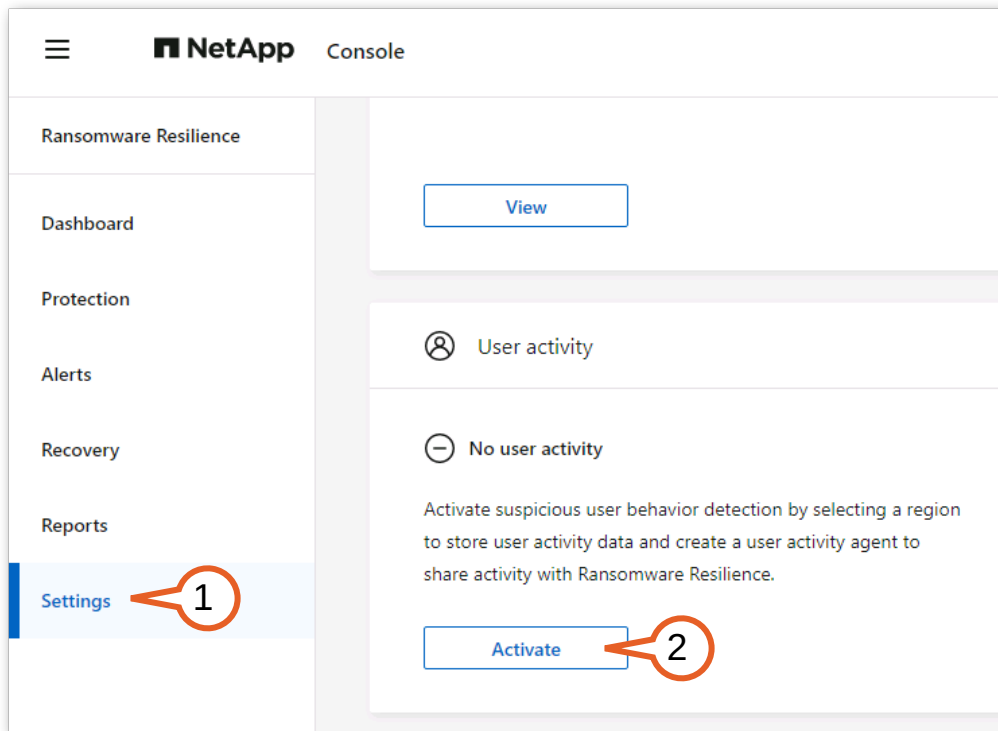


Figure 2-31:

3. The User activity agent requires that some additional data be stored in the cloud. These storage costs are covered by your subscription.
You can pick your preferred cloud provider. Click **Amazon Web Services**.
4. Under Region, select **us-east-1**.

5. Click **Next**.

Region

Select a cloud provider and region to store your user activity data. This can't be changed later. [Learn more](#)

Microsoft Azure

Amazon Web Services

Region

us-east-1

Cancel Next

Figure 2-32:

6. The User activity agent is different from the Console Agent. This agent will send data specifically for Ransomware Resilience.

Enter the following:

- User activity agent name: **agent1**
- Select Console agent: **rt1*****-onpremAgent**
- VM DNS name or IP address: **192.168.0.61**

User activity agent

Configure the user activity agent in your environment to allow it to send data to Ransomware Resilience. Adding a user activity agent can take several minutes.

[Get help](#)

[Linux versions supported](#) [Minimum server requirements](#)

User activity agent name

agent1

Select a Console agent located near the user activity agent to minimize latency when transmitting activity to Ransomware Resilience.

Console agent

rt1*****-onpremAgent

Provide the VM executable environment with "root" access for collectors in this user activity agent.

VM DNS name or IP address

192.168.0.61

Figure 2-33:

7. For VM SSH key, open the **ssh.txt** file on the desktop and copy its contents.

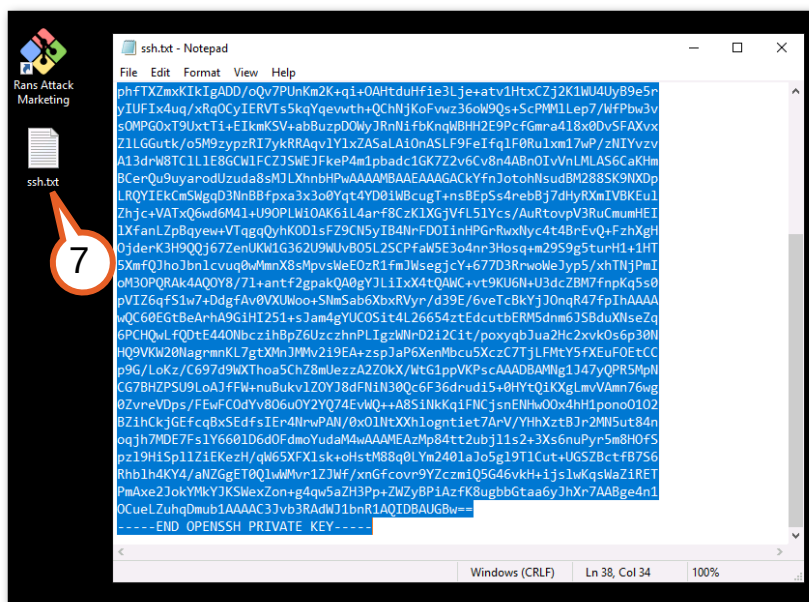


Figure 2-34:

8. Paste the contents of ssh.txt in the VM SSH key field.
9. Click Next.

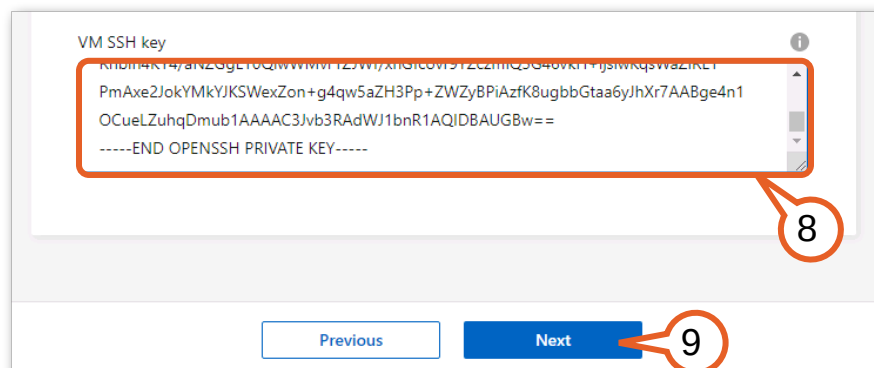


Figure 2-35:

10. Click **Activate**.

Review

Region

Amazon Web Services Cloud provider	us-east-1 Region
---------------------------------------	---------------------

User activity agent

agent1 Name	rt1- Console agent	192.168.0.61 VM DNS or IP address
----------------	-----------------------	--------------------------------------

[Previous](#) [Activate](#)

Figure 2-36:

11. Under **User activity**, you should see the status “Activating”. Wait approximately three minutes.

User activity

No user activity

Activate suspicious user behavior detection by selecting a region to store user activity data and create a user activity agent to share activity with Ransomware Resilience.

Activating

Figure 2-37:

12. When activation is complete, click **Manage**.

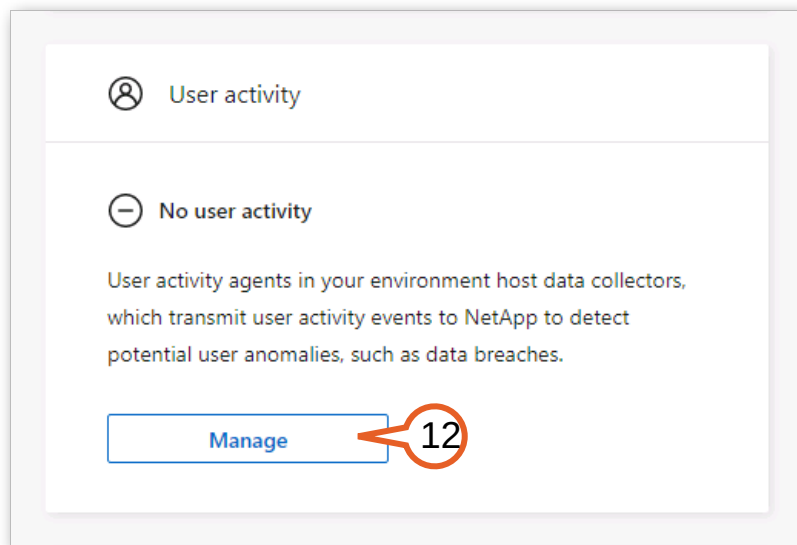


Figure 2-38:

13. Click the **User activity agents** tab.
14. Confirm that agent1 was created and has a status of “Connected”. If not, you may need to wait 3-5 minutes before you see this status.

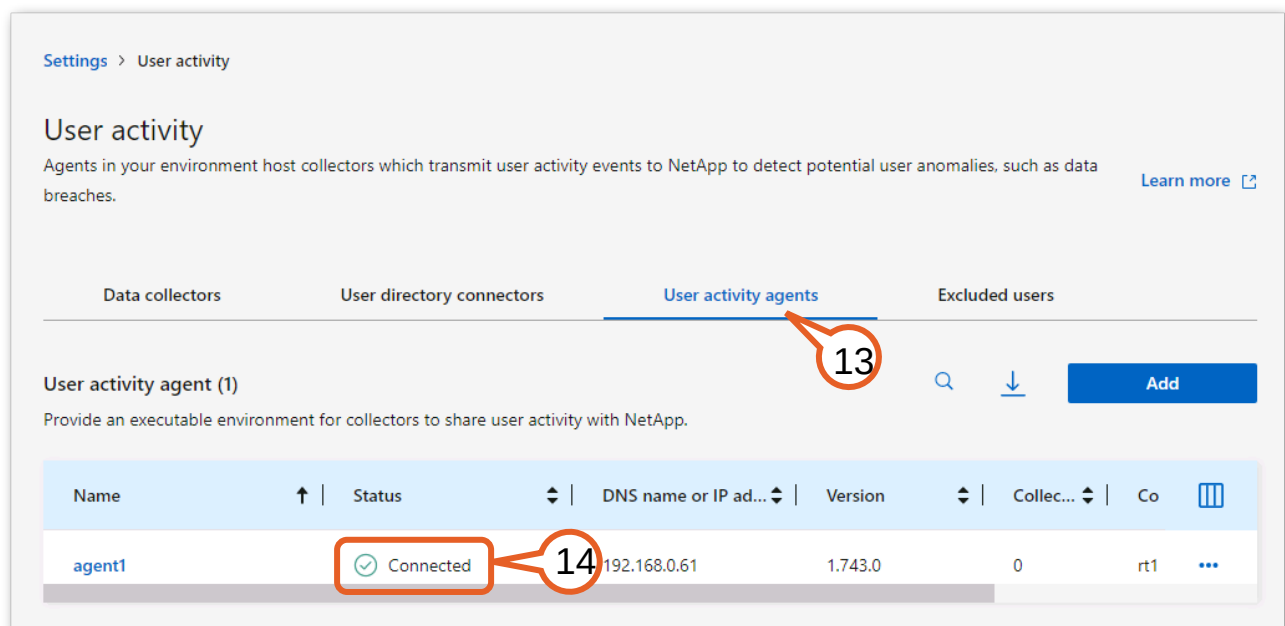


Figure 2-39:

15. Click **User directory connectors**.

16. Click **Add**.

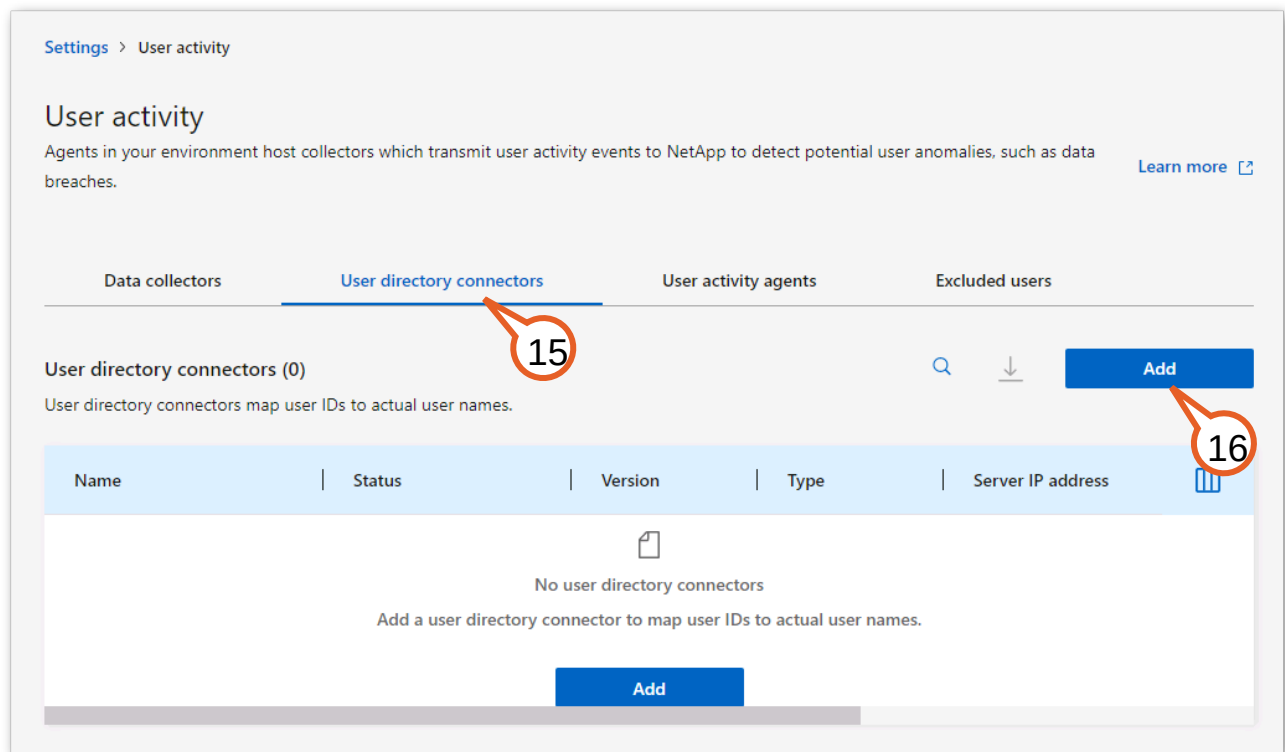


Figure 2-40:

17. Click **Connection**.

18. Enter the following information in the fields available.

- Name: **dc1**
- User Activity Agent: **agent1**
- Server IP or DNS name: **192.168.0.253**
- Forest name or search name: **demo.netapp.com**
- Bind DN: **CN=Administrator,CN=Users,DC=demo,DC=netapp,DC=com**
- Bind Password: **Netapp1!**

You can leave the remaining fields with their default values.

User directory

Connect to your user directories to identify specific users performing potentially suspicious behavior. [Get help](#)

Connection 17

Name: dc1

User directory type: Active Directory

User activity agent: agent1

Server IP or DNS name: 192.168.0.253

Forest name or search name: demo.netapp.com

Bind DN: CN=Administrator,CN=Users,DC=demo,DC=netapp,DC=com 18

Bind password:

Protocol: LDAP

Port: 389

Figure 2-41:

19. Click **Add**.

The screenshot shows a configuration window with two expandable sections: 'Attribute mapping' and 'Advanced'. The 'Attribute mapping' section shows 'Required attributes: 4' and 'Optional attributes: 0'. The 'Advanced' section shows a search query: '((objectClass=posixAccount)(objectCategory=person)(objectClass=user))'. At the bottom, there are 'Cancel' and 'Add' buttons. The 'Add' button is highlighted with a red circle and the number 19.

Figure 2-42:

20. You should see the User directory connector “dc1” in a state of “Initializing”. Wait about two minutes for the connector to initialize, showing a status of “Running”.

The screenshot shows the 'User directory connectors' tab in the interface. It displays a table with columns: Name, Status, Version, Type, and Server IP address. The connector 'dc1' is listed with a status of 'Running' (indicated by a green checkmark icon). The 'Status' column is highlighted with a red circle and the number 20.

Name	Status	Version	Type	Server IP address
dc1	Running	1.376.0	Active Directory	192.168.0.253

Figure 2-43:

With your User activity agent created and configured, you are ready to create your protection strategy which puts all of the NetApp Ransomware Resilience protection capabilities together.

2.2.4 Enable Protection

The most important elements of NetApp Ransomware Resilience are the protection strategies. These strategies bundle together several policies, making it easy to apply layered protections very quickly to your workloads. In this activity, you examine a protection strategy and then apply this strategy to a single workload and a group of workloads.

1. Following the previous activity, you should still be in the console for NetApp Ransomware Resilience. Click **Protection**.

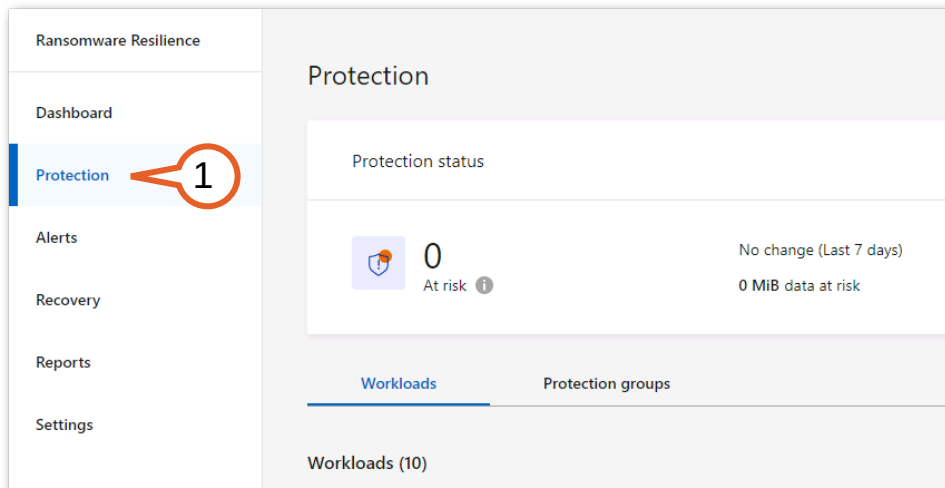


Figure 2-44:

2. Notice that next to each workload there is a column for Type that tells you if it's a database, a file share, or even data sitting on Block storage in a LUN.
3. For any workload that is protected, you will see a status value in any of the three columns:
 - Encryption detection status, displays when attacks to ONTAP happen that are reported from Autonomous Ransomware Protection with AI (ARP/AI).
 - Suspected user behavior detection, which is a data breach capability when a user activity agent is configured.
 - Block suspicious file extensions, which utilizes FPolicy in ONTAP.
4. Click **Manage protection strategies**.

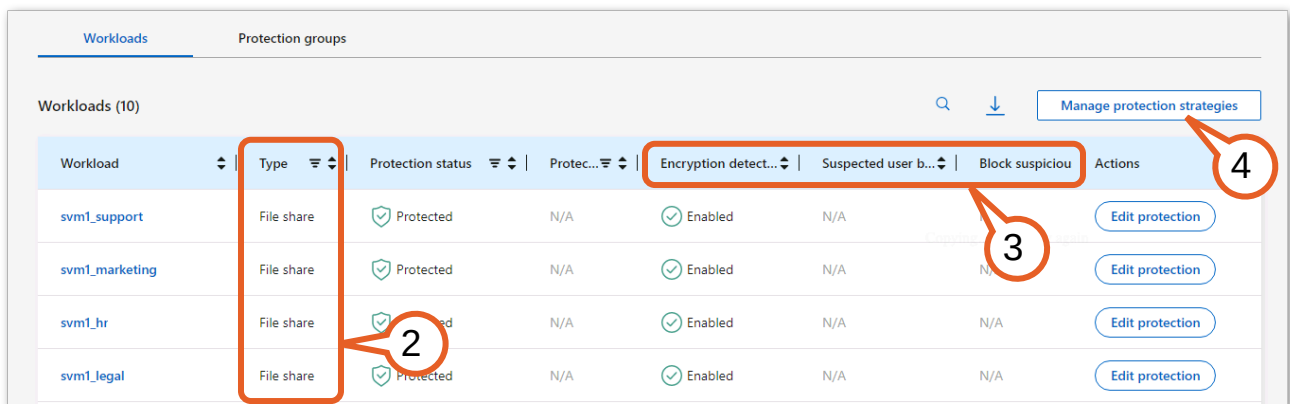
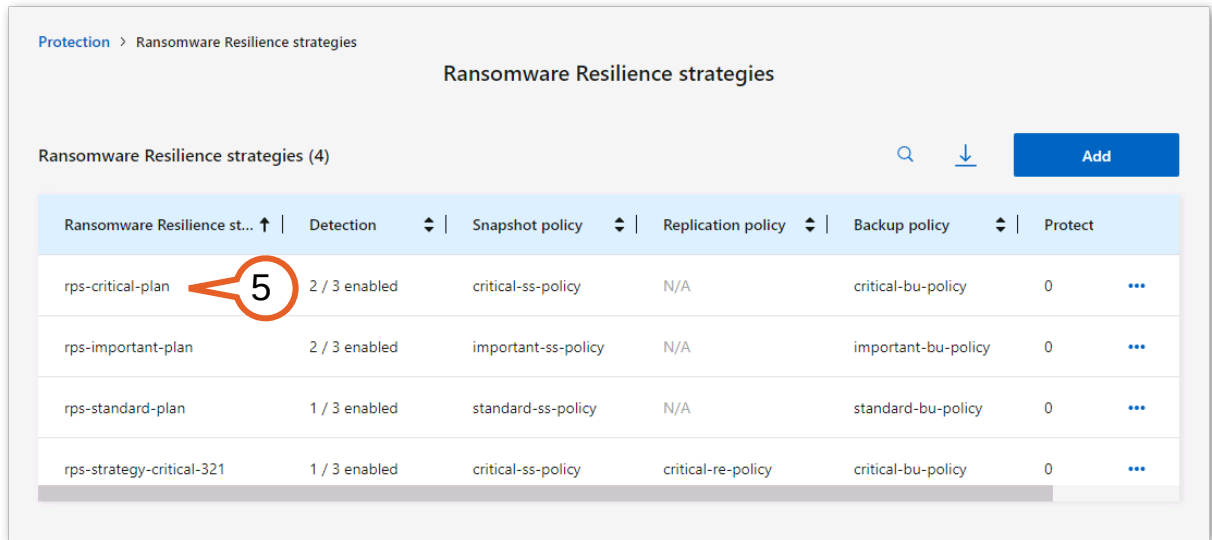


Figure 2-45:

- On the Ransomware protection strategies page, you can review or create ransomware protection strategies. To better understand ransomware protection strategies, click **rps-critical-plan** to expand its row.



Protection > Ransomware Resilience strategies

Ransomware Resilience strategies

Ransomware Resilience strategies (4) 🔍 ⬇️ Add

Ransomware Resilience st... ↑	Detection ⬆️	Snapshot policy ⬆️	Replication policy ⬆️	Backup policy ⬆️	Protect
rps-critical-plan	2 / 3 enabled	critical-ss-policy	N/A	critical-bu-policy	0 ...
rps-important-plan	2 / 3 enabled	important-ss-policy	N/A	important-bu-policy	0 ...
rps-standard-plan	1 / 3 enabled	standard-ss-policy	N/A	standard-bu-policy	0 ...
rps-strategy-critical-321	1 / 3 enabled	critical-ss-policy	critical-re-policy	critical-bu-policy	0 ...

Figure 2-46:

- Here you can see that this strategy consists of three policies:

- Detection policy:** this includes two major components, and in the case of the rps-critical-plan strategy, both are enabled:
 - “Encryption detection” is powered by Autonomous Ransomware Protection (ARP) with AI in ONTAP. ARP with AI is an entirely on-box solution that detects anomalies across all user activity. It uses a sophisticated AI model that is trained on common workloads and continues to learn from your NAS workloads. When ARP detects a ransomware attack, it provides you with valuable information about the attack, and takes a proactive snapshot that is designed to give you the enhanced recovery options.
 - “Block suspicious file extensions” uses FPolicy. FPolicy is an ONTAP capability that provides a granular, file-based event notification interface that enables either ONTAP or partner applications connected to your storage systems to monitor and set file access permissions. This FPolicy implementation is run internally within ONTAP and is designed to block common ransomware file extensions.

For more information about ARP and FPolicy, see the lab “Protection and Recovery From Ransomware”.

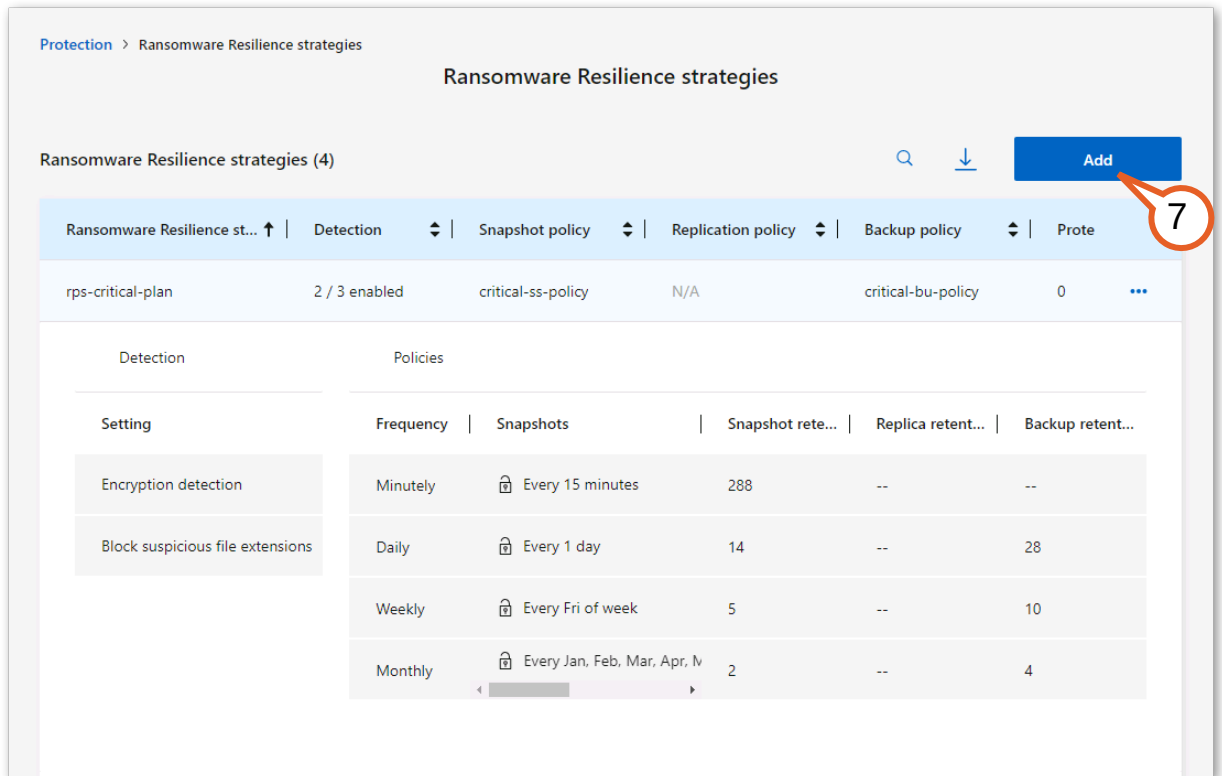
- Snapshot policy:** this policy automatically takes snapshot copies. NetApp snapshot copies are space-efficient and high performance, meaning you can retain many snapshots without consuming much storage, and you can clone or restore in place from a snapshot very quickly. Snapshot policies take snapshots on one or more configured schedules and determine how many of these snapshot copies are retained for each of the schedules. You can also enable snapshot locking (not enabled for this strategy), which will prevent the snapshots from being deleted during a retention period. For the rps-critical-plan, snapshots are taken every 15 minutes and kept for three days, while weekly and monthly snapshot copies are retained for months.
- Backup policy:** this policy takes a backup of the workload storage on one or more schedules. Like the snapshot policy, you provide a retention count to control how many backups are retained. These backups are either stored on object storage managed by NetApp Ransomware Resilience, or they correspond to your workload’s existing backups using either NetApp Backup and Recovery,

SnapCenter, or SnapCenter Plug-in for VMware vSphere. The rps-critical-plan takes a backup every day, retaining those for four weeks, while keeping weekly and monthly backups for up to four months.

Takeaway: Storage admins are provided with recommended protection policies based on workload importance to provide the most cost-efficient protection strategy, and ensure that the most critical workloads get the level of protection that they need.

In addition to the pre-existing protection strategies, you can create your own strategy including custom detection, snapshot, and backup policies.

7. For your workloads you will create a new Ransomware Resilience strategy to use instead of using an existing one. Click **Add**.



Protection > Ransomware Resilience strategies

Ransomware Resilience strategies

Ransomware Resilience strategies (4)

Search Download Add

Ransomware Resilience st...	Detection	Snapshot policy	Replication policy	Backup policy	Prote
rps-critical-plan	2 / 3 enabled	critical-ss-policy	N/A	critical-bu-policy	0

Detection	Policies				
Setting	Frequency	Snapshots	Snapshot rete...	Replica retent...	Backup retent...
Encryption detection	Minutely	Every 15 minutes	288	--	--
Block suspicious file extensions	Daily	Every 1 day	14	--	28
	Weekly	Every Fri of week	5	--	10
	Monthly	Every Jan, Feb, Mar, Apr, Iv	2	--	4

Figure 2-47:

8. Enter **strategy1**.

- Expand the Detection settings.

Add Ransomware Resilience strategy

Ransomware Resilience strategy name: Copy from existing Ransomware Resilience strategy: No Ransomware Resilience strategy selected [Select](#)

Detection 1 / 3 enabled

Snapshot Policy Action required

Replication policy Depends on snapshot policy selection

Backup policy Depends on snapshot policy selection

Figure 2-48:

- Under Detection, toggle all three detection methods to enable them.



Note: Suspicious user behavior detection is available to enable because the previous lab activity activated the user activity agent, which provides enhanced alerts and more detailed detection based on user actions.

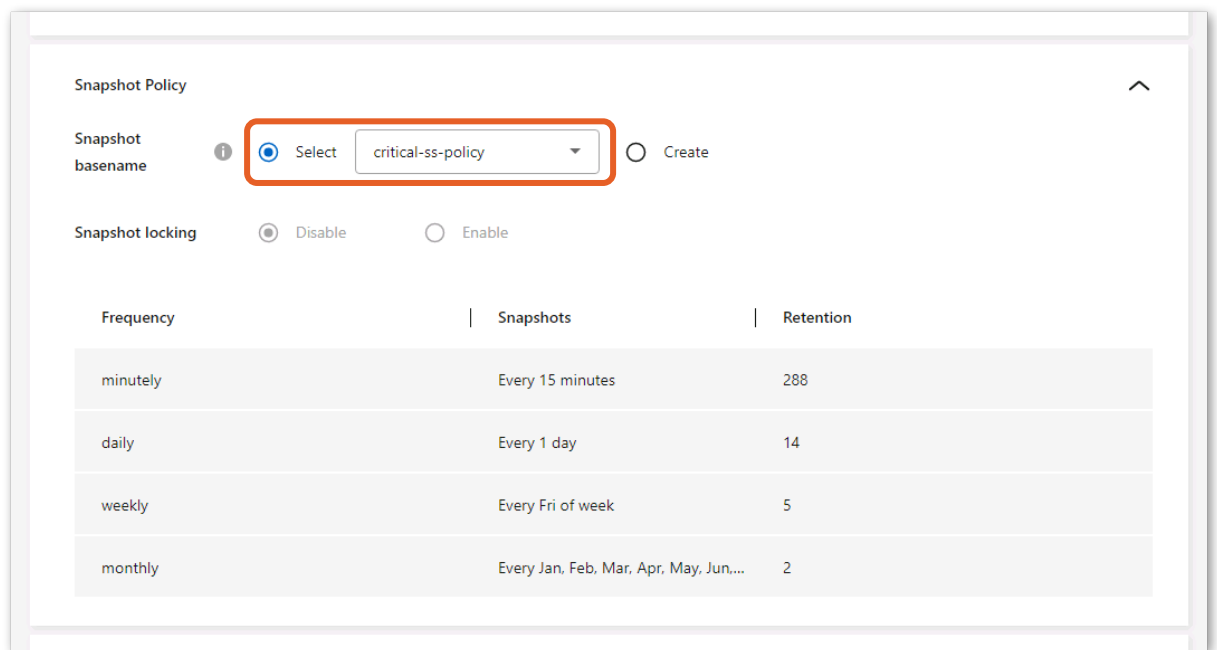
Detection

Enable detection settings to detect ransomware and malicious behavior.

Enable	Setting
☒	Encryption detection
☒	Block suspicious file extensions
☒	Suspicious user behavior detection

Figure 2-49:

11. Under Snapshot policy, click **Select** and select **critical-ss-policy**.



Snapshot Policy

Snapshot basename ☒ Select critical-ss-policy ☐ Create

Snapshot locking ☒ Disable ☐ Enable

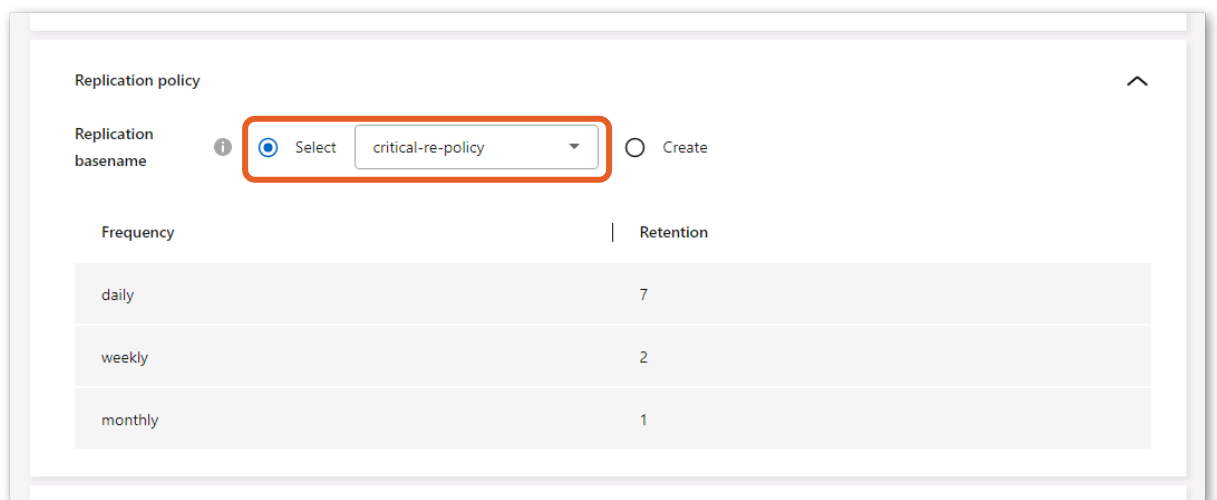
Frequency	Snapshots	Retention
minutely	Every 15 minutes	288
daily	Every 1 day	14
weekly	Every Fri of week	5
monthly	Every Jan, Feb, Mar, Apr, May, Jun,...	2

Figure 2-50:



Note: You do have the option to create a new snapshot policy and set the frequency, retention settings, and schedule right from here.

12. Under Replication policy, click **Select** and select **critical-re-policy**.



Replication policy

Replication basename ☒ Select critical-re-policy ☐ Create

Frequency	Retention
daily	7
weekly	2
monthly	1

Figure 2-51:

13. Under Backup policy, click **Select** and select **critical-bu-policy**.

14. Click **Add**.

Backup policy

Backup basename ? ☒ Select critical-bu-policy ☐ Create

Frequency	Retention
daily	28
weekly	10
monthly	4

Figure 2-52:

15. Click **Protection** near the top of the page to navigate back to the protection tab.

Protection > Ransomware Resilience strategies

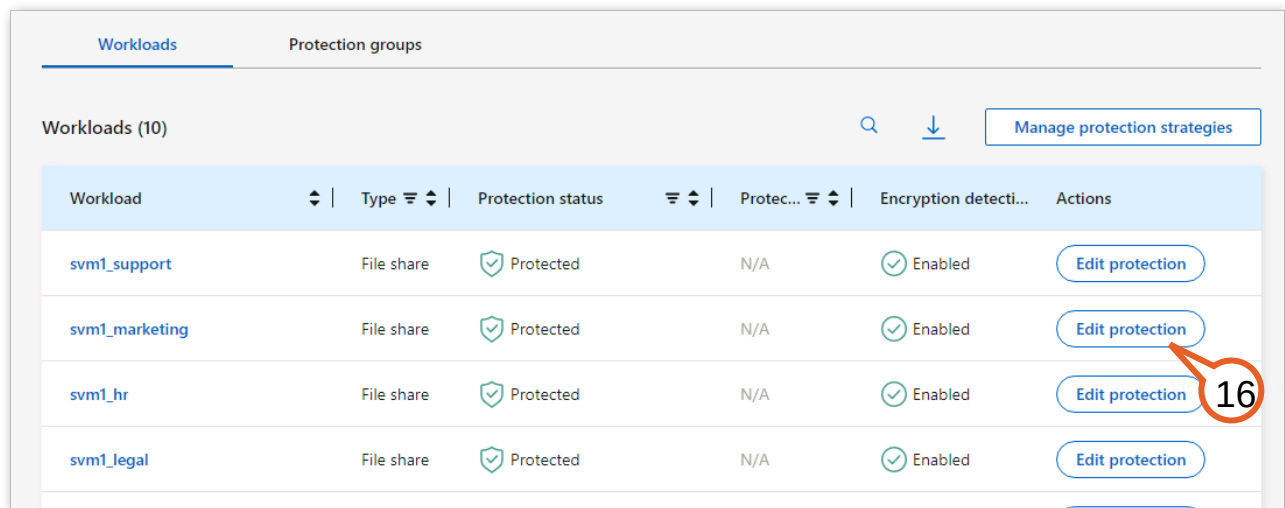
Ransomware Resilience strategies

Ransomware Resilience strategies (5) 🔍 ⬇️

Ransomware Resilience st...	Detection	Snapshot policy	Replication policy	Backup policy	Protect
rps-critical-plan	2 / 3 enabled	critical-ss-policy	N/A	critical-bu-policy	0
rps-important-plan	2 / 3 enabled	important-ss-policy	N/A	important-bu-policy	0
rps-standard-plan	1 / 3 enabled	standard-ss-policy	N/A	standard-bu-policy	0
rps-strategy-critical-321	1 / 3 enabled	critical-ss-policy	critical-re-policy	critical-bu-policy	0
strategy1	3 / 3 enabled	critical-ss-policy	critical-re-policy	critical-bu-policy	0

Figure 2-53:

16. Locate the svm1_marketing workload, and click **Edit Protection** on the right.



Workload	Type	Protection status	Protec...	Encryption detecti...	Actions
svm1_support	File share	Protected	N/A	Enabled	Edit protection
svm1_marketing	File share	Protected	N/A	Enabled	Edit protection
svm1_hr	File share	Protected	N/A	Enabled	Edit protection
svm1_legal	File share	Protected	N/A	Enabled	Edit protection

Figure 2-54:

17. Because svm1_marketing already has a snapshot policy, you will have to instruct Ransomware Resilience to take over managing its protection.

Check the box for **Replace policies managed by Backup and Recovery**.



Protection
Select how to protect the workload **svm1_marketing**

☐ This workload already has policies managed by Backup and Recovery.

☒ Replace policies managed by Backup and Recovery

Figure 2-55:

18. Select **strategy1**.

19. Click **Next**.

Ransomware Resilience strategy ↑ | Detection ⇅ | Snapshot policy ⇅ | Replication policy ⇅ | Backup policy ⇅ |

☐	rps-critical-plan	Recommended	2 / 3 enabled	critical-ss-policy	N/A
☐	rps-important-plan		2 / 3 enabled	important-ss-policy	N/A
☐	rps-standard-plan		1 / 3 enabled	standard-ss-policy	N/A
☐	rps-strategy-critical-321		1 / 3 enabled	critical-ss-policy	critical-re-policy
☒	strategy1		3 / 3 enabled	critical-ss-policy	critical-re-policy

18 on

Policies

Settings	Frequency	Snapshots	Snapshot ret...	Replica reten...	Backup reten...
Encryption detection	Minutely	Every 15 minutes	288	--	--
Block suspicious file extensions	Daily	Every 1 day	14	7	28
Suspicious user behavior detection	Weekly	Every Fri of week	5	2	10
	Monthly	Every Jan, Feb, Mar, Apr, May, Jun, Jul, Aug, Sep, Oct, Nov, Dec	2	1	4

Snapshot locking Disabled

Cancel Next 19

Figure 2-56:

20. On the Replication destinations page, select **Cluster3** under Destination system.
21. Select **svm3** for Destination storage VM.

22. Click **Next**.

Replication destinations

Select the replication destination for the workload, **svm1_marketing**

Workload (1)

Workload	Destination system	Destination storage VM
svm1_marketing	Cluster1	svm1

Previous Next

Figure 2-57:

23. On the Backup destination page, select **sg1**.
24. Click **Next**.

Backup destinations

Select how to protect the workload **svm1_marketing**

System (1)

System	Volumes	Backup lock	Backup destination
Cluster1	1	None	sg1

Previous Next

Figure 2-58:

25. Since you enabled the User activity detection you are asked to identify the user activity agent to use. Select **agent1**.
26. Click the **Test connection** option to make sure the user activity agent can communicate with ONTAP.

27. After about a minute, the test should complete, indicating success with a green check mark.
Click **Next**.

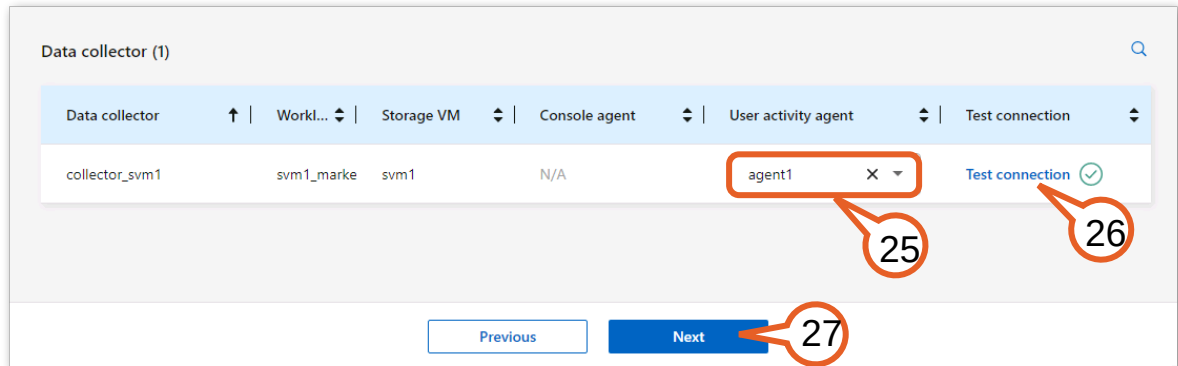


Figure 2-59:

28. After reviewing the summary, click **Protect**.

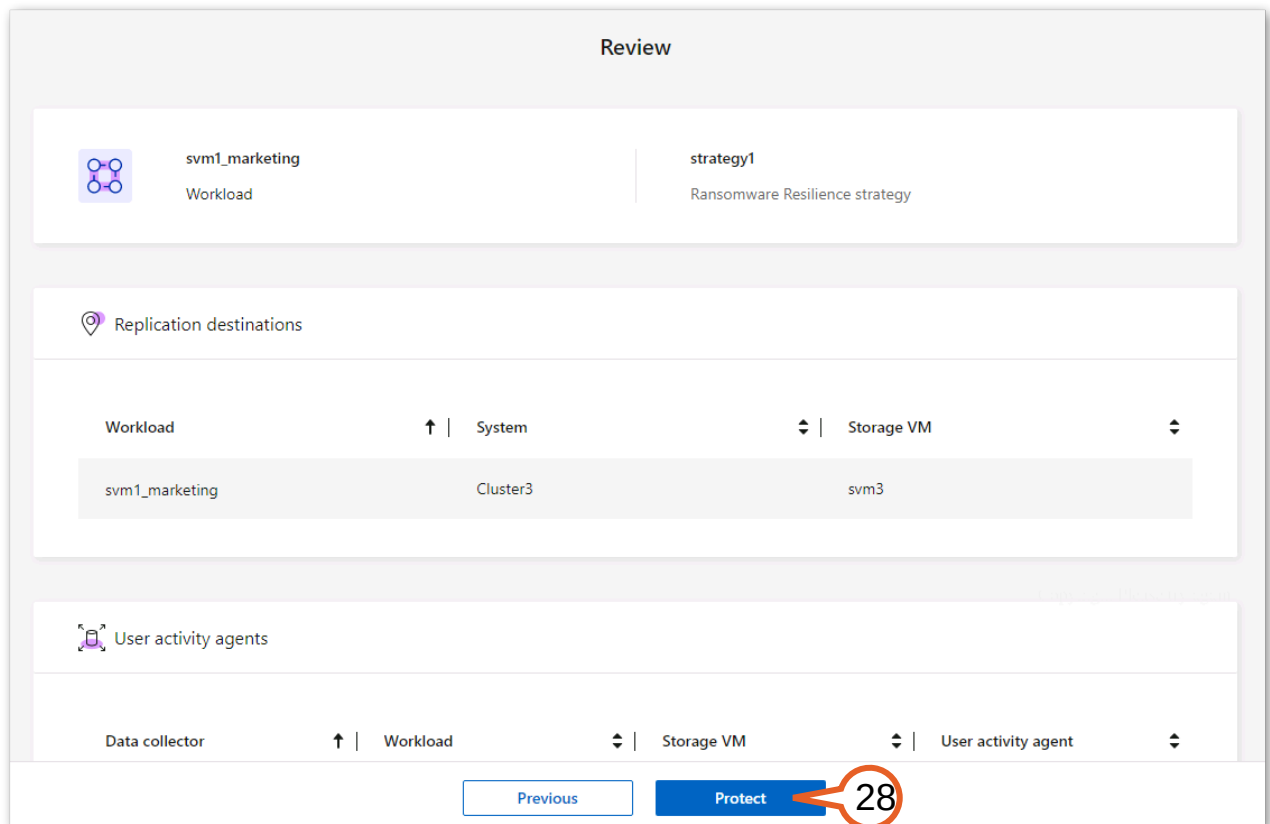


Figure 2-60:

29. It will take a few minutes for the workload to be protected. Feel free to click on the **Refresh** icon at the top right.

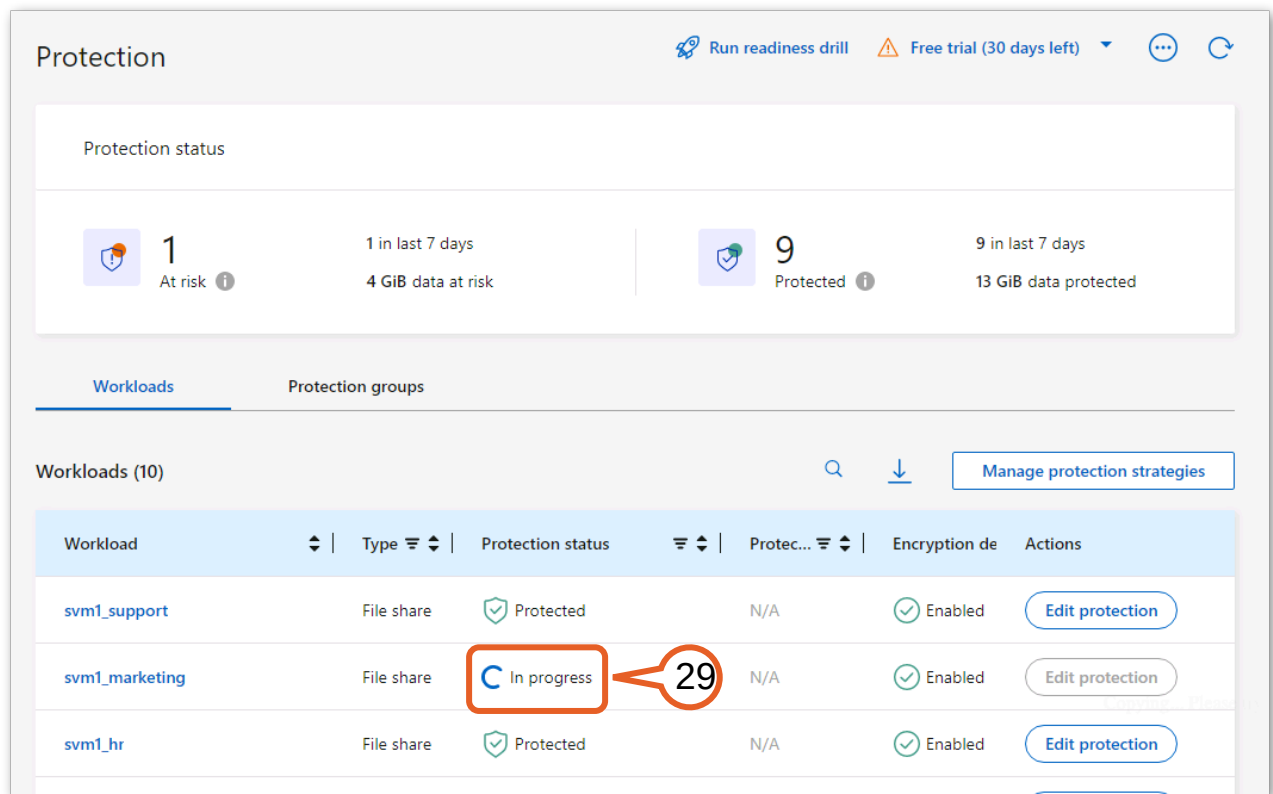


Figure 2-61:

30. Once complete, the workload shows it is “Protected” under the Protection status.

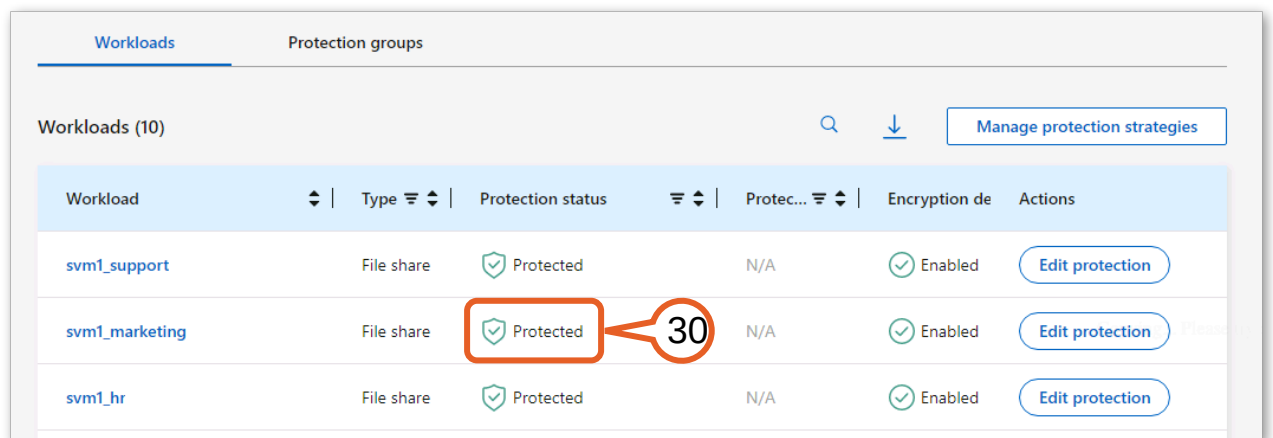


Figure 2-62:

31. Click the **Protection groups** tab.

32. Protection groups allow you to group multiple workloads together and then apply a protection strategy to all workloads in that grouping. Click **Add**.

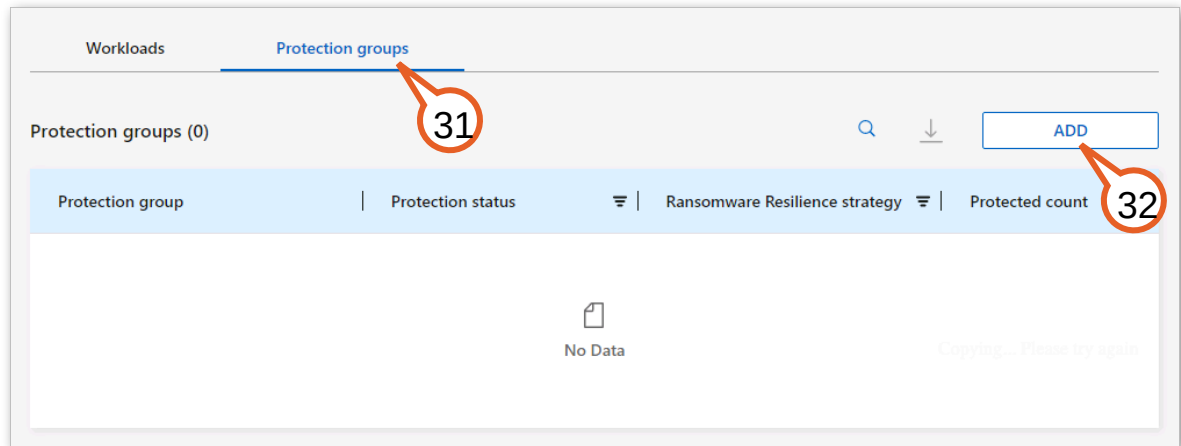
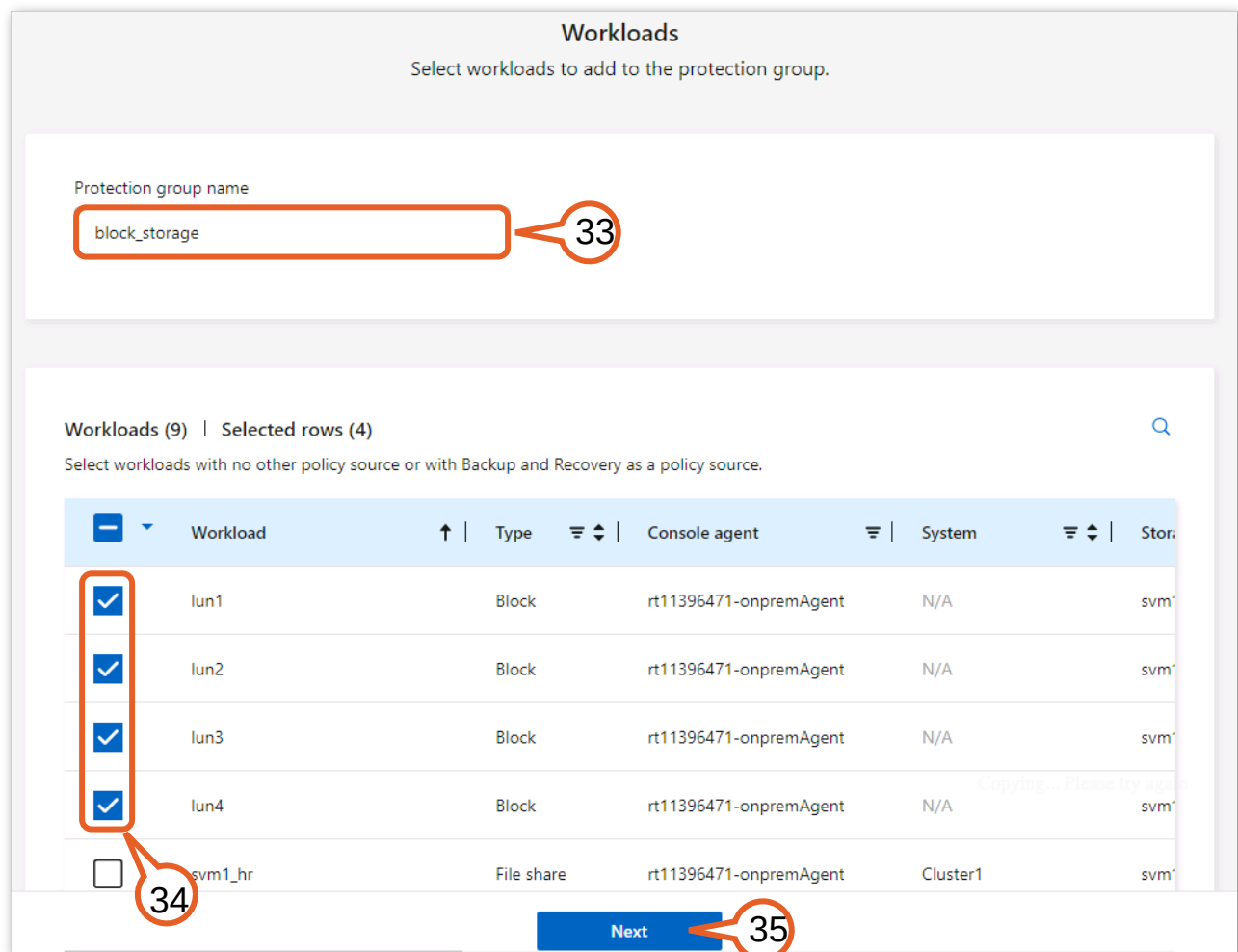


Figure 2-63:

33. Enter **block_storage** for the Protection group name.
34. Select all four **lun*** workloads.

35. Click **Next**.



Workloads
Select workloads to add to the protection group.

Protection group name
block_storage 33

Workloads (9) | Selected rows (4)

Select workloads with no other policy source or with Backup and Recovery as a policy source.

	Workload	Type	Console agent	System	Storage
☒	lun1	Block	rt11396471-onpremAgent	N/A	svm1
☒	lun2	Block	rt11396471-onpremAgent	N/A	svm1
☒	lun3	Block	rt11396471-onpremAgent	N/A	svm1
☒	lun4	Block	rt11396471-onpremAgent	N/A	svm1
☐	svm1_hr	File share	rt11396471-onpremAgent	Cluster1	svm1

34 35

Next

Figure 2-64:

36. Select **rps-standard-plan** which is one of the default strategies.

37. Click **Next**.

Ransomware Resilience strategies (5) | Selected rows (1)

Ransomware Resilience strategy↑	Detection	Snapshot policy	Replication policy	Backup policy
☐ rps-critical-plan	2 / 3 enabled	critical-ss-policy	N/A	critical-bu-policy
☐ rps-important-plan	2 / 3 enabled	important-ss-policy	N/A	important-bu-policy
☒ rps-standard-plan	1 / 3 enabled	standard-ss-policy	N/A	standard-bu-policy

36

Policies

Settings	Frequency	Snapshots	Snapshot re...	Replica rete...	Backup rete...
Encryption detection	Hourly	Every 1 hours	72	--	--
	Daily	Every 1 day	14	--	14

Previous Next

37

Figure 2-65:

38. On the Review page, select **Protect**.

Workloads

Workload	Type	Console agent	Importance	Priv
lun1	Block	rt11396471-onpremAgent	Standard	n/a
lun2	Block	rt11396471-onpremAgent	Standard	n/a
lun3	Block	rt11396471-onpremAgent	Standard	n/a
lun4	Block	rt11396471-onpremAgent	Standard	n/a

1 - 4 of 4 << < 1 > >>

Previous Protect

38

Figure 2-66:

Takeaway: Comprehensive ransomware protection strategies can be applied to workloads and groups of workloads with very few clicks, reducing the need for complicated and time-consuming volume-by-volume configurations

across multiple environments. With protection enabled, storage is monitored for any future ransomware activity and recovery options are ensured with snapshot copies, volume backups on object storage, or application consistent backups with SnapCenter.

One of your workloads is now protected from the threat of ransomware or data breach.



Important: From here you may choose which type of incident you would like to simulate and resolve. To avoid unexpected results in your lab, please choose one type of security breach:

- To simulate a ransomware attack, proceed to 2.3, [Option 1: Respond to a Ransomware Attack](#).
- To simulate a data exfiltration incident, proceed to 2.4, [Option 2: Respond to a Data Breach](#)

2.3 Option 1: Respond to a Ransomware Attack

When a ransomware attack is suspected, admins need to be immediately alerted so that they can quickly respond. To respond appropriately, they also need as much information as possible to understand the nature and scope of the attack. Additionally, after the attack is neutralized, admins need fast and easy workload recovery options with minimized data-loss. NetApp Ransomware Resilience provides the tools necessary to satisfy these requirements.

In the following lab activities, you use NetApp Ransomware Resilience to detect and recover from an attack.

Lab Activities
Detect a Ransomware Attack
Recover from a Ransomware Attack

2.3.1 Detect a Ransomware Attack

This activity demonstrates the ONTAP anti-ransomware feature in action. First, you assume the role of a cyber criminal by running a script against the svm1_marketing volume that encrypts all of its files, mimicking a ransomware attack. After that you resume your duties as a storage administrator and examine the alert of a suspected ransomware attack within NetApp Ransomware Resilience.

1. From the Jumpshot desktop, double-click the **Ransomware Attack**.

2. This script encrypts all of the files in the svm1_marketing file share.

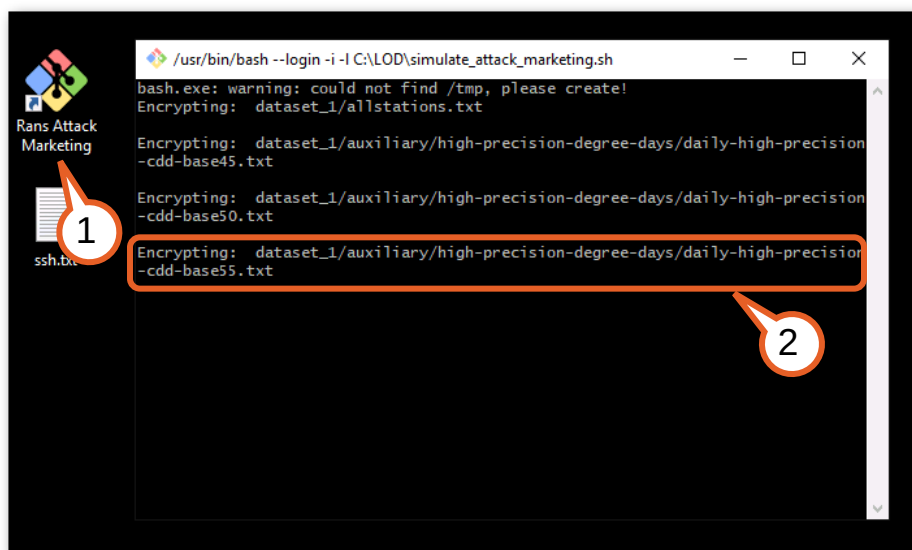


Figure 2-67:



Tip: If you would like to see the file damage caused by the attack, you may navigate to the **svm1_marketing** network drive within File Explorer on your Jumpshot. Initial file encryption starts in the "M:\dataset_1\auxiliary\high-precision-degree-days:" directory.

3. Now you will resume your role as storage administrator. Return to Ransomware Resilience and click the **Alerts** tab. Wait for an alert to appear in this view.



Important: In this scenario, it can take up to 15 minutes for Ransomware Resilience to detect the ransomware activity.

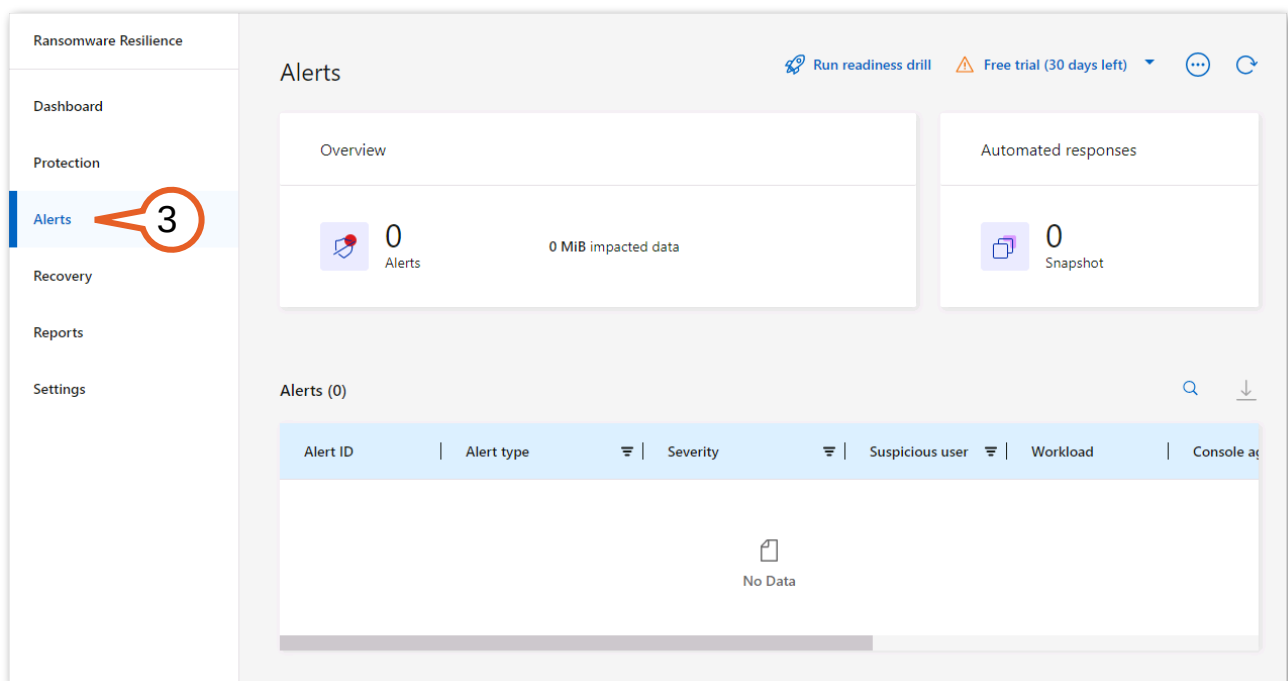


Figure 2-68:

4. When the alert appears, notice there is also a red badge on the NetApp Console notifications icon. Click the **notifications icon**.



Note: In some cases you may need to refresh the in-lab web browser to see the notification.

5. The potential ransomware attack alerts are surfaced here and can also be emailed to administrators of this NetApp Console organization (depending on your NetApp Console notification settings).

Click **outside the notification panel** to dismiss it.



Tip: NetApp Ransomware Resilience also lets you connect to a security information and event management (SIEM) system for threat reporting. You can configure this by clicking the menu icon at the top-right of the page and navigating to Settings, and then clicking **Connect** under SIEM connection.

Additionally, within ONTAP when ARP detects abnormal volume activity, it generates an Event Management System (EMS) notification which can be forwarded directly to an email address, a syslog server, or a Simple Management Network Protocol (SNMP) trap host.

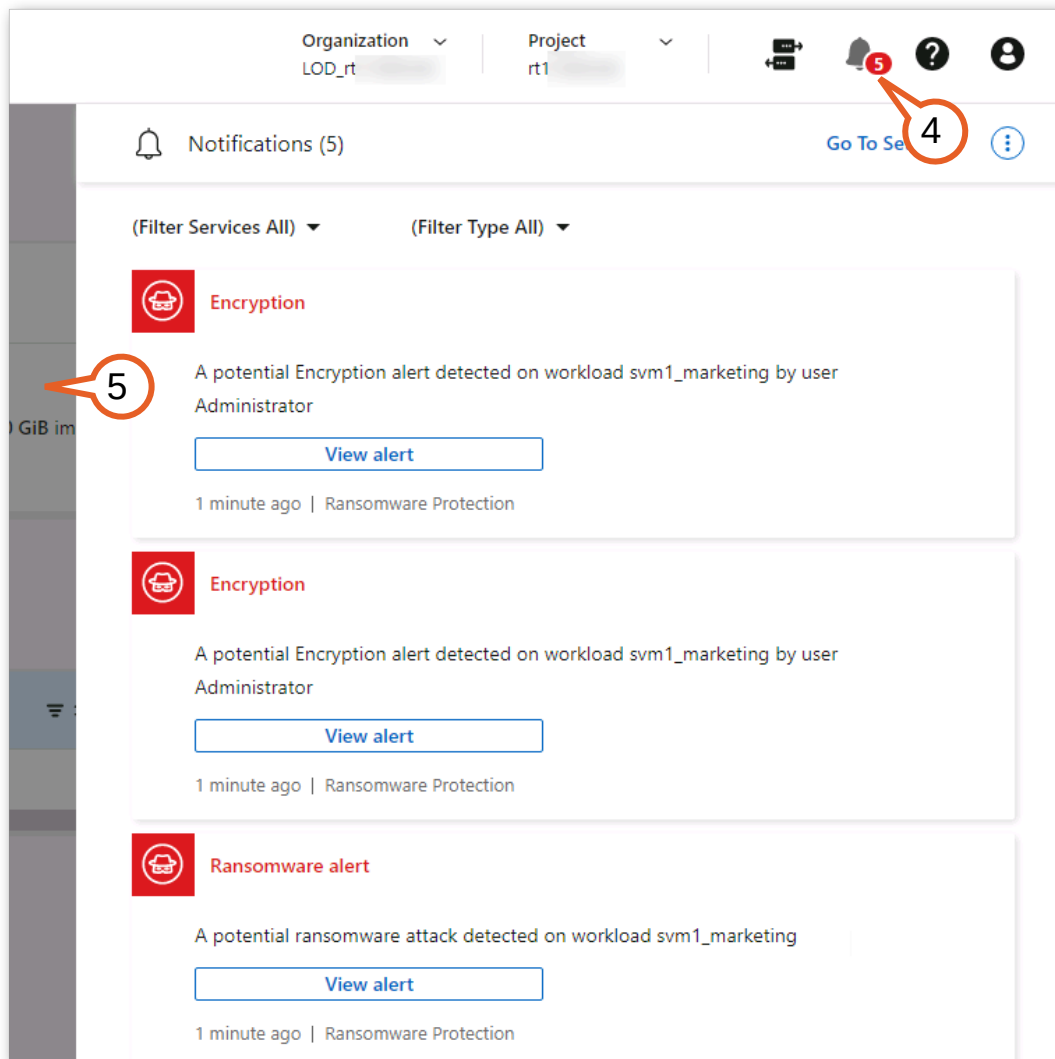


Figure 2-69:

Takeaway: NetApp Ransomware Resilience (along with Autonomous Ransomware Protection in ONTAP) provides you with timely notifications of suspicious activity.

6. In the alert table, there are multiple possible alert types including:

- Encryption
- Data Breach
- Data Destruction
- Suspicious user behavior

You may see multiple alerts from your simulated ransomware attack because in addition to the file encryption, a significant number of read and deleted operations were done which resembles a data breach.

From the Alerts table, locate the alert with “Encryption” as the Alert type and “svm1_marketing” as the Workload and click the **alert ID**.

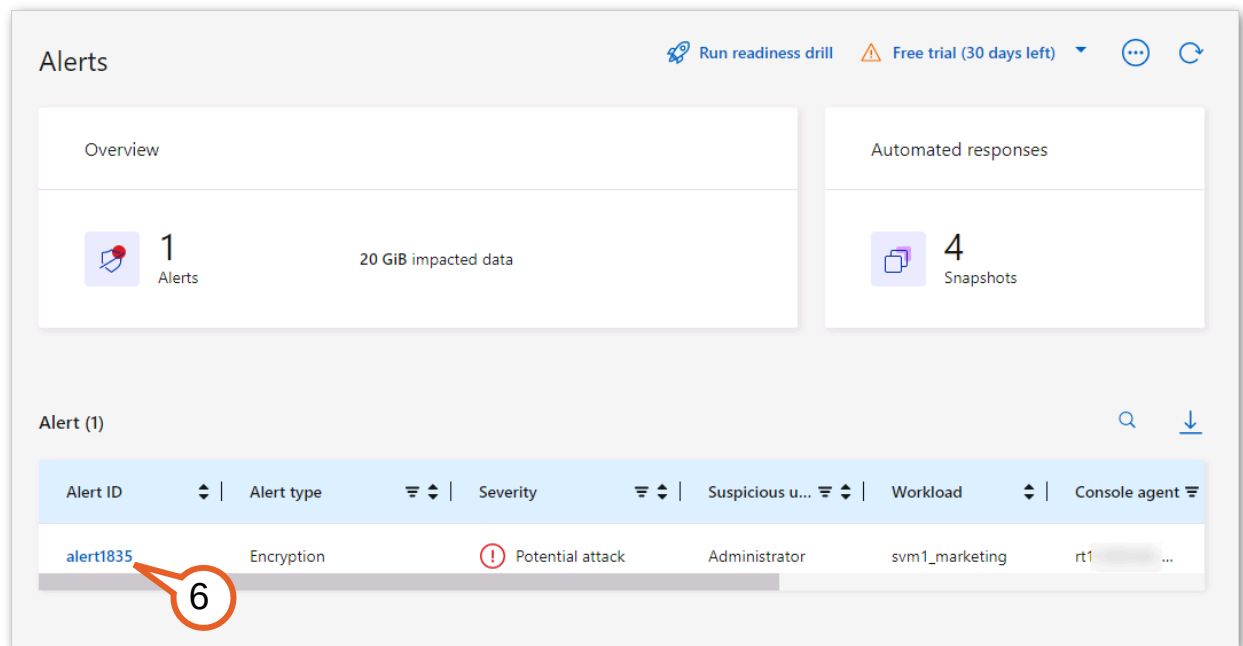


Figure 2-70:

7. The alert provides information about where the attack occurred, when it was first detected, and how much data is impacted. Each alert can have multiple incidents on different volumes with different statuses. From the incident table you can find out more granular information about the attack as well as dismiss non-threatening activity.

The service provides information called “evidence” about what caused the alert to be issued, such as the following:

- File extensions were created or changed.
- File creation occurred and increased by a listed percentage.
- File deletion occurred and increased by a listed percentage.

In this case the evidence presented is likely the new file extension used by the attack script.

The alert will be classified as one of two types:

- **Potential attack:** An alert occurs when Autonomous Ransomware Protection detects a new extension and the occurrence is repeated more than 20 times in the last 24 hours (default behavior).
- **Warning:** A warning occurs when there is a new extension but with less repetition, high entropy, or an unusual increase in IOPS.

8. Click the **incident ID** in the Incident table.

The screenshot displays the 'Alerts > alert1835' page. At the top, it shows 'alert1835' with 'Impacted workloads: svm1_marketing' and a 'Mark restore needed' button. Below this is a summary bar (callout 7) containing four items: a lightning bolt icon with '1 Potential attack', a folder icon with '341 Impacted files', a database icon with '20 GiB Impacted data', and a clock icon with 'March 26, 2026, 2:3... First detected'. Below the summary bar is an 'Incident (1)' section with a table. The table has columns: Incident ID, Volume, Storage VM, System, Severity, and a link icon. The first row (callout 8) contains: a checkbox, the incident ID 'inc5286', the volume 'svm1_marketing', the storage VM 'svm1', the system 'Cluster1', and the severity 'Potential attack' with a red warning icon.

Incident ID	Volume	Storage VM	System	Severity
inc5286	svm1_marketing	svm1	Cluster1	Potential attack

Figure 2-71:

9. Here you can see more information about the incident at the volume level.

The Suspicious activity tab is selected by default and displays information about impacted files, identified extensions, and additional file activity.

10. Click **All activity**.

Alerts > alert1835 > inc5286

inc5286

Impacted workloads: svm1_marketing

Administrator
Block
Suspicious user

New Status

Potential attack Severity

Encryption Type

March 26, 2026, ...
Detection time

Suspicious activity All activity 10

Review the suspicious file activity that triggered this potential encryption attack. 9

Incoming data

Entropy of incoming data
Detected Not determined (Surge detection in progress)
Expected 3756 KiB / min

File activity

706
Suspected files

Suspect file extensions
.lckd123

Files (10/706)

Path	File
/dataset_1/	allstations.txt.lckd123
/dataset_1/auxiliary/high-...	daily-high-precision-cdd-...

Figure 2-72:

11. This tab shows you file activity over time, allowing you to visually assess how significant the unusual activity is compared to activity before the incident.

12. At the top you may see some information about any identified suspicious users associated with the attack. This feature is enabled with User behavior monitoring that you enabled in a previous activity.

Click **Block** to block this users access to the file system.

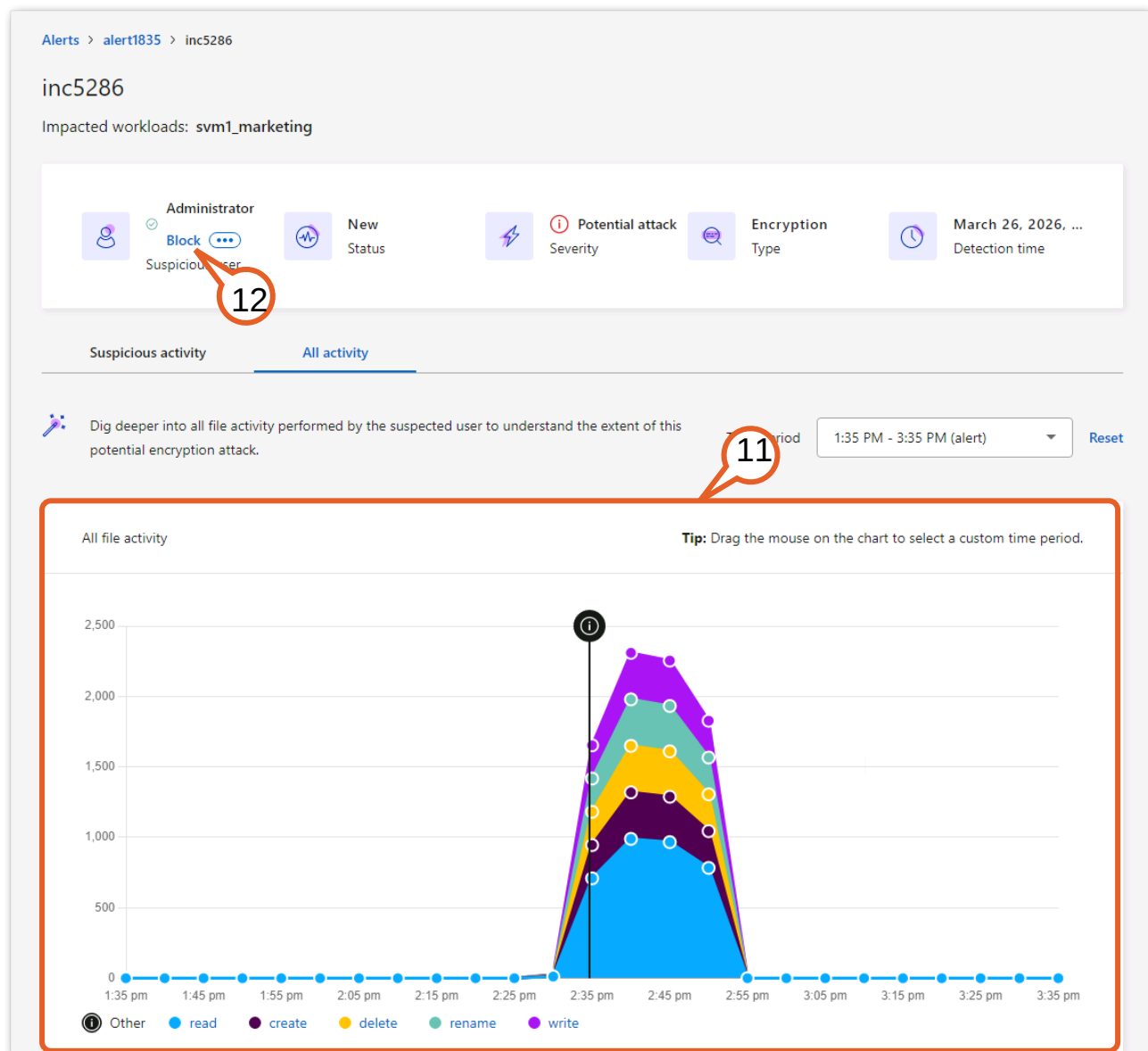


Figure 2-73:

13. In the confirmation pop-up, click **Block**.

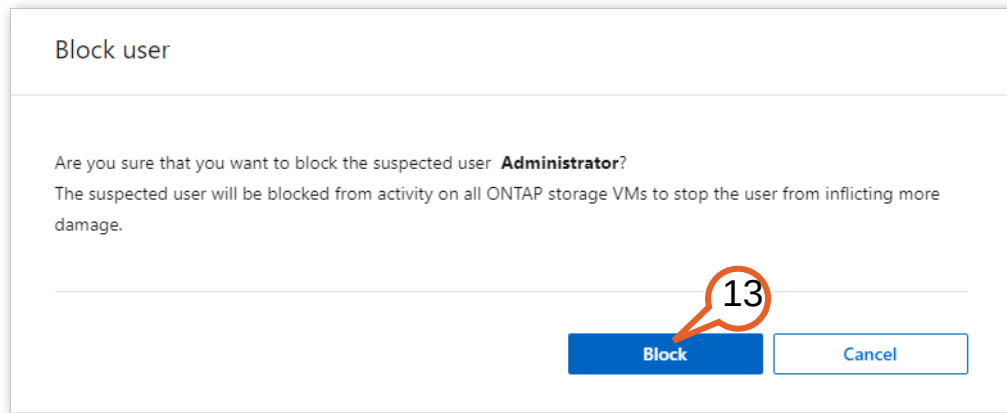


Figure 2-74:

14. Return to the attack script window. Within a minute, you should see the attack produce “Encryption failed” messages as access to the filesystem has been revoked.

Click the **close** button at the top-right of the attack script window.

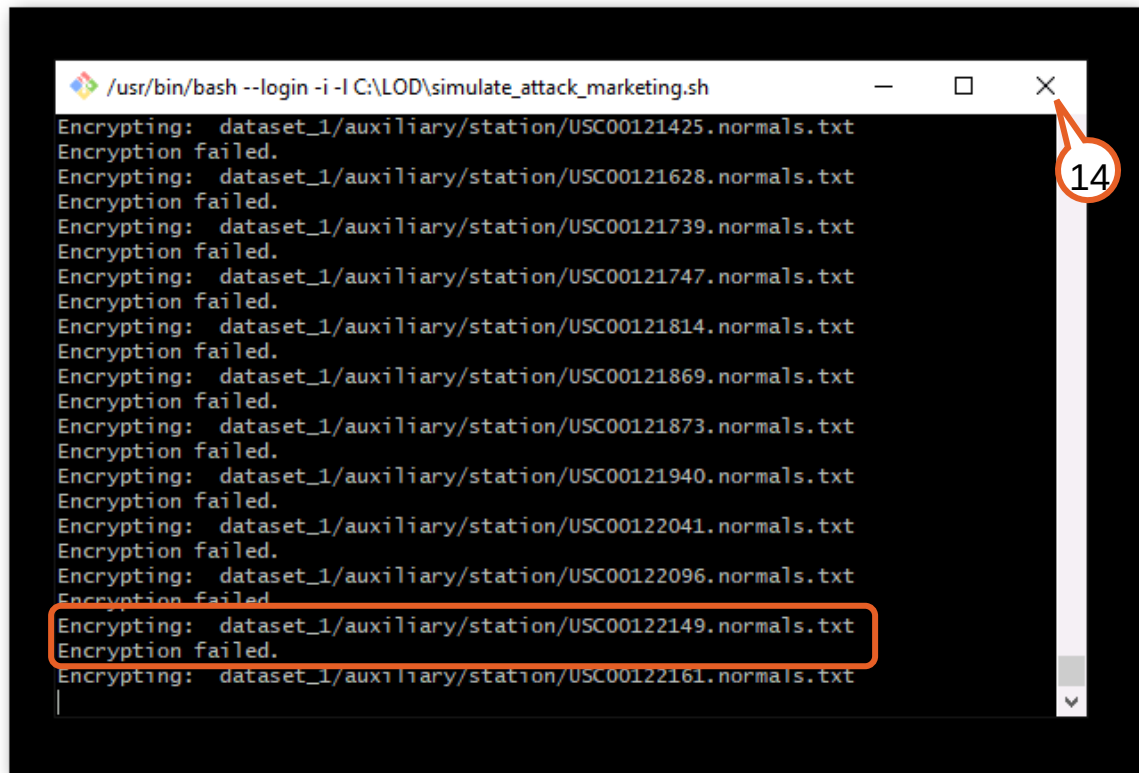


Figure 2-75:

15. If prompted to confirm, click **OK**.

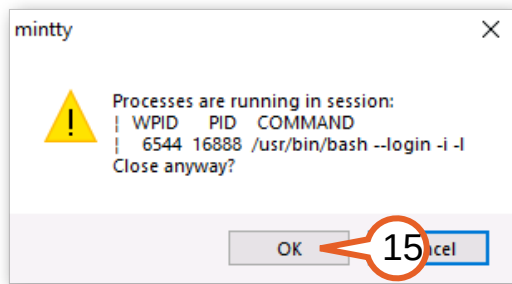


Figure 2-76:

16. Return to Ransomware Resilience and click the **alert ID** at the top-left to go back to the alert view.

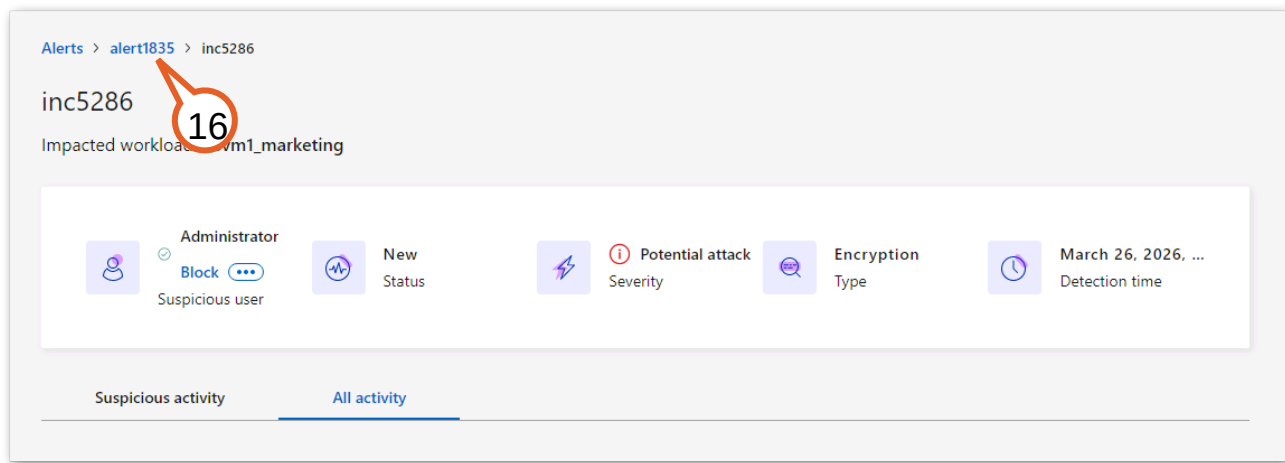


Figure 2-77:

17. Now that you have investigated the alert and blocked the suspicious user, you are confident that you have stopped a real ransomware attack. The next step is to recover any damage from the attack. Click **Mark restore needed**. This will move this workload to the Recovery tab.

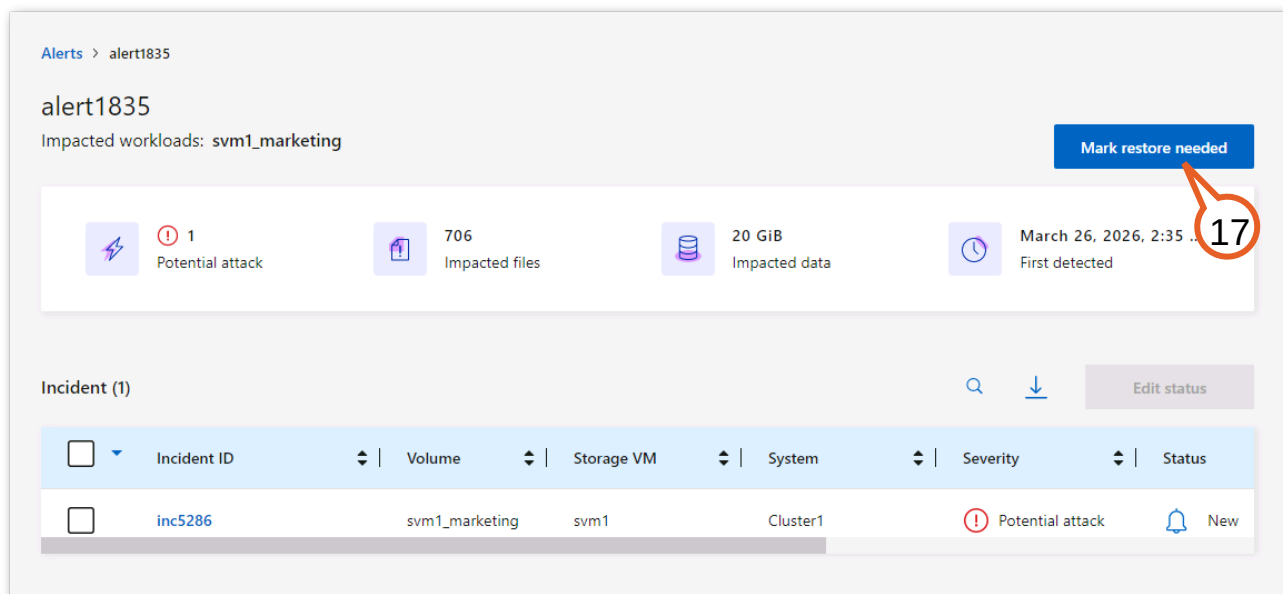


Figure 2-78:

18. Click **Mark restore needed** to confirm.

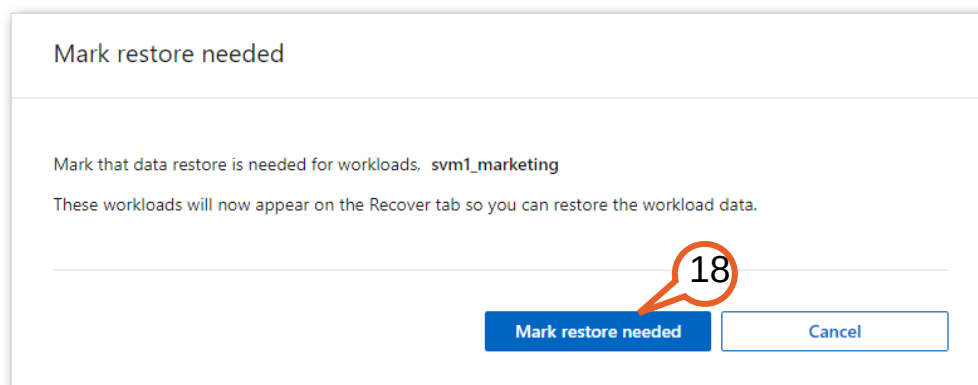


Figure 2-79:

Takeaway: NetApp Ransomware Resilience helps you easily respond to a suspected ransomware attack by providing you with all the information you need, and providing an easy-to-manage triage workflow for your workloads.

With the suspicious user identified and blocked you are ready to begin recovering from the attack. In the next activity, you will use the Ransomware Resilience recovery workflow to restore the encrypted data.

2.3.2 Recover from a Ransomware Attack

When you are alerted of a possible ransomware attack you need to be able to react quickly. You want to recover from the attack by restoring data to a state as close as possible to its state immediately prior to the attack, thus minimizing data losses. With each passing minute, the cost of the disruption to your company's operations increases, so you need to stop the attack and recover the data as quickly as possible.

In this activity, you walk through the recovery process for your workload and take advantage of the recovery points offered as a result of your protection strategy.

1. To get started on the recovery, click the **Recovery** tab in NetApp Ransomware Resilience.

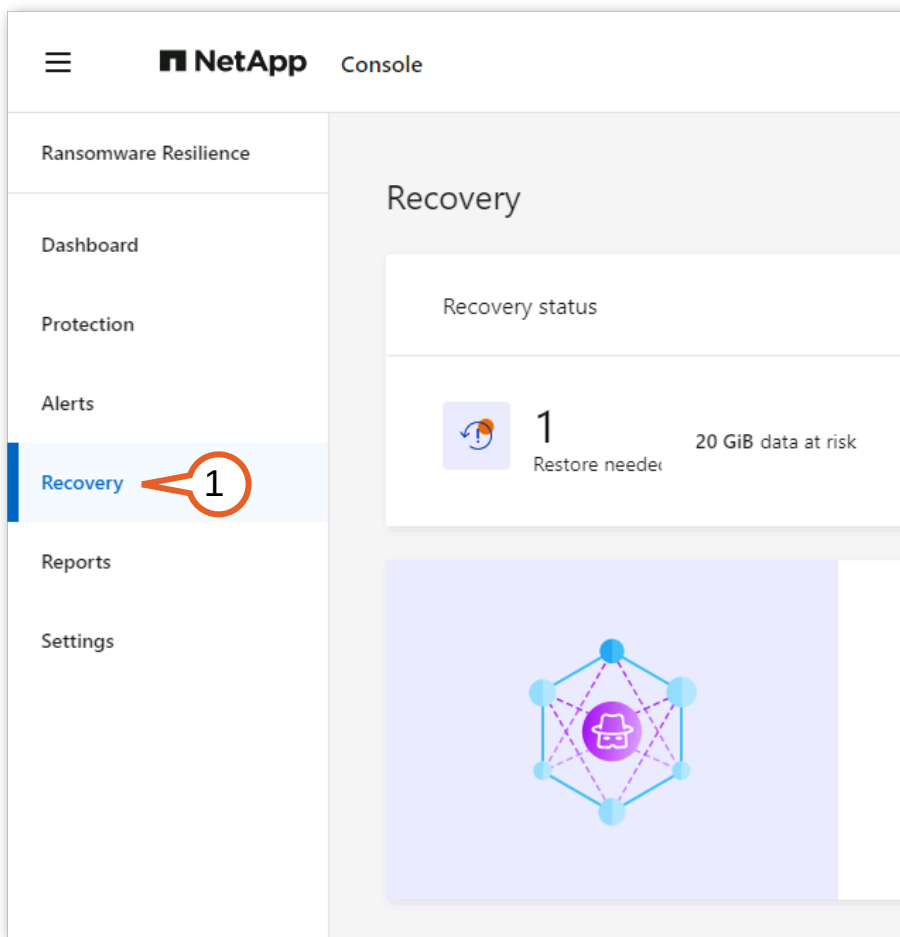


Figure 2-80:

2. On the Recovery page you can see a list of workloads that need recovery. This workflow can be helpful to manage scenarios where recovery is needed across several workloads. Click **Restore** on the right side for the svm1_marketing workload.

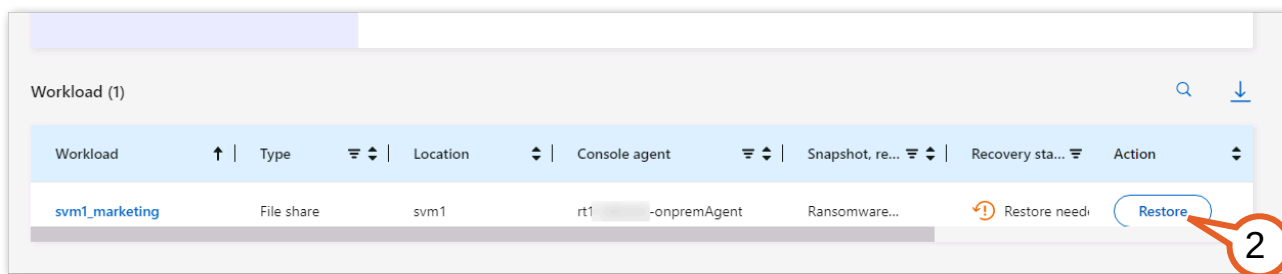


Figure 2-81:

3. You have multiple restore scopes you can choose from, including:

- All volumes
- By volume
- By file

Restoring by file allows you to restore specific files that you select, or from a list that you upload, or from a list of impacted files that is generated automatically by Ransomware Resilience.

Select **By volume** under Restore scope.

4. Now select **svm1_marketing** from the volume list on the left. If multiple volumes were involved in this workload you could work through each recovery one-by-one.
5. Click the **downward arrow** to the right of the Source row to expand it.

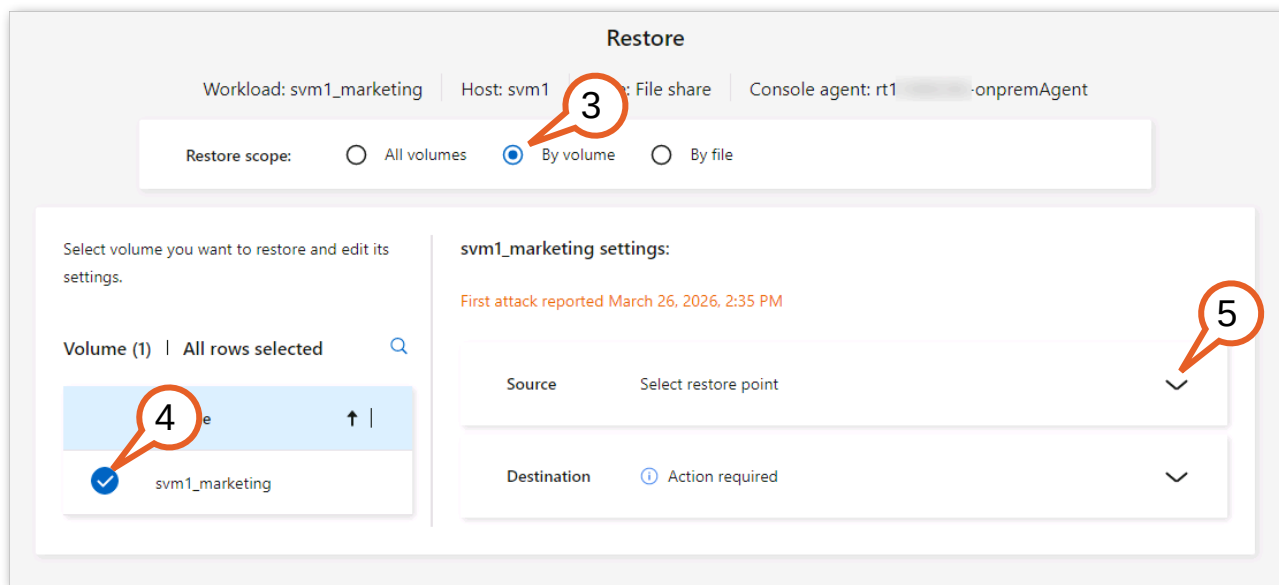


Figure 2-82:

6. You will see a date and time at the top telling you when the first attack was reported. This allows you to find the most suitable restore point to prevent the least amount of data loss.

Locate a source restore point from before the time of the attack that is of type “Snapshot”, and that is closest to the attack time. Snapshots stored locally on Cluster1 will provide the fastest restore time, but in the event snapshots are compromised the other methods are available because of your protection strategy.



Note: You may need to scroll through pages of restore options if enough time has passed since the simulated attack in your lab.

svm1_marketing settings:

First attack reported March 26, 2026, 2:35 PM

Source

Volumes (14) | Selected rows (1)

	Restore point	Type	Date	
☐	Anti_ransomware_attack_b...	Snapshot	March 26, 2026, 2:48 PM	4
☐	rps-critical-ss-policy-...	Snapshot	March 26, 2026, 2:45 PM	4
☒	rps-critical-ss-policy-...	Snapshot	March 26, 2026, 2:30 PM	4
☐	6 snapmirror.b0345dbc-169...	Snapshot	March 26, 2026, 2:23 PM	4
☐	snapmirror.0c7e589b-36d...	Snapshot	March 26, 2026, 2:23 PM	4
☐	20min.2026-03-26_1420	Snapshot	March 26, 2026, 2:20 PM	4
☐	Anti_ransomware_periodic...	Snapshot	March 26, 2026, 2:17 PM	4
☐	20min.2026-03-26_1400	Snapshot	March 26, 2026, 2:00 PM	4
☐	20min.2026-03-26_1340	Snapshot	March 26, 2026, 1:40 PM	4
☐	Anti_ransomware_periodic...	Snapshot	March 26, 2026, 10:05 AM	4

1 - 10 of 14 << < 1 > >>

Figure 2-83:

7. Expand the Destination section and select **Clone to new volume**.
8. In the Volume name field, enter **marketing_clone**,

9. Click **Save**.

svm1_marketing settings:

First attack reported March 26, 2026, 2:35 PM

Source

Restore point: rps-critical-ss-polic... | Type: Snapshot |

Date: March 26, 2026, 2:30 PM

Destination

Restore options

☐ Original Location

☒ Clone to new volume

Volume name

marketing_clone

Save

Figure 2-84:

10. Click **Next**.

Save

Next

Figure 2-85:

11. Review your selections and click **Restore**.

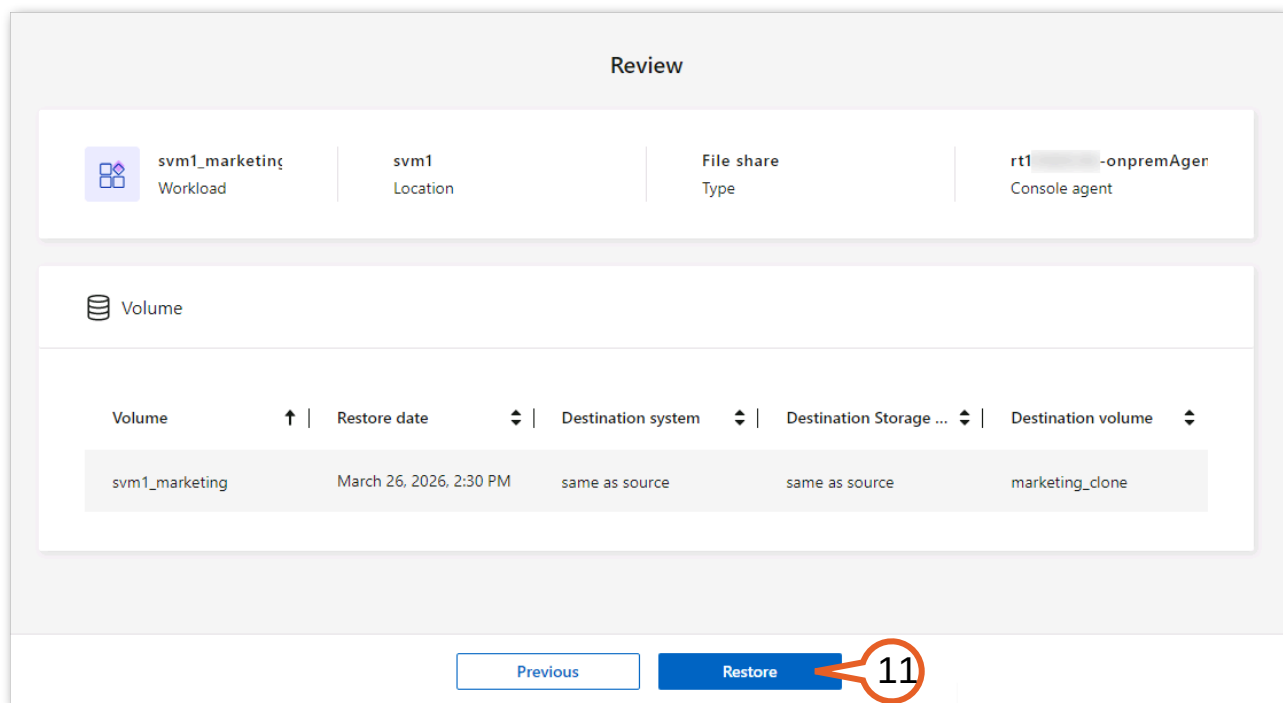


Figure 2-86:

12. You are returned to the Recovery tab where you can monitor the progress of the restore.



Note: You will likely need to horizontally scroll the table to see the “Recovery status” column.

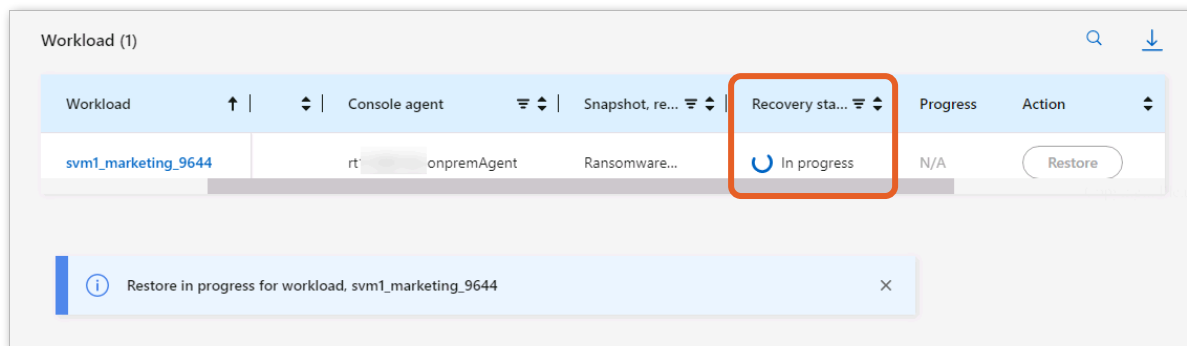


Figure 2-87:

- Once complete, the Recovery status will change to “Restored”.



Tip: If you would like, you can examine the restored contents of the cloned volume by navigating to **Storage > Management** in the NetApp Console menu. From there you can navigate to System Manager for Cluster1, mount the marketing_clone volume and then create a SMB share for the volume.

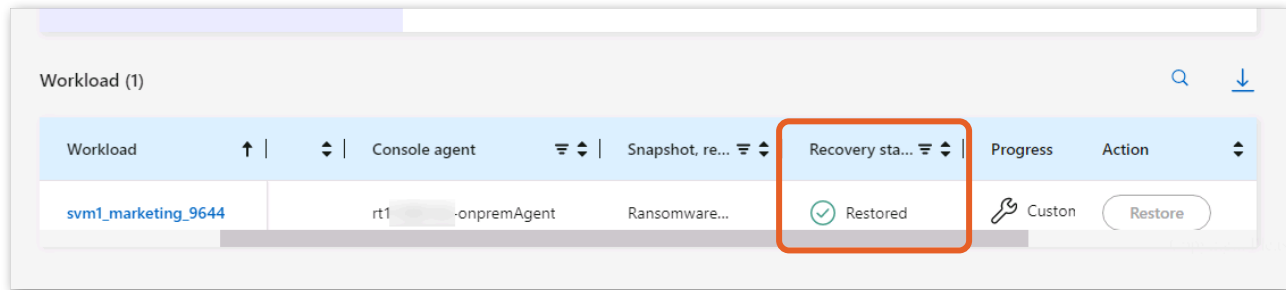


Figure 2-88:

- Click **Reports** from the menu on the left.
- With the situation resolved, you might now want to generate reports of the affected workloads and actions taken during this incident. NetApp Ransomware Resilience can automatically generate CSV reports of protected workloads, alerts, and recovery operations.

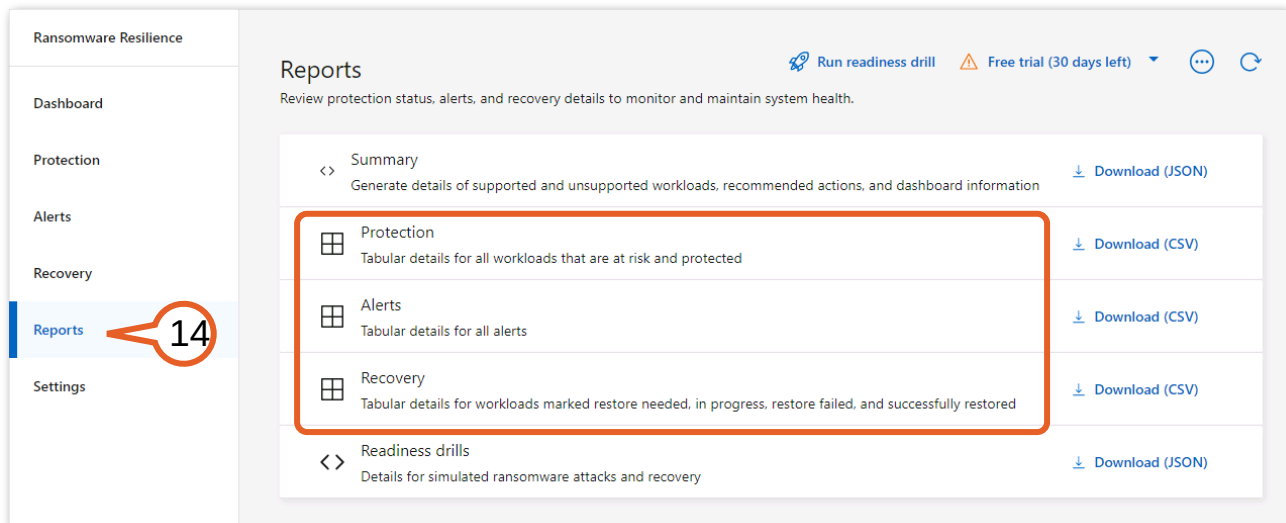


Figure 2-89:

Takeaway: With data restored by taking advantage of robust recovery options provided by NetApp Ransomware Resilience, you were able to fix the damage caused by ransomware in minutes.

2.4 Option 2: Respond to a Data Breach

NetApp Ransomware Resilience includes data breach detection, an AI-driven capability that identifies anomalous user and file system behaviors that are early indicators of potential data exfiltration and thus a breach attempt. Upon identification, Ransomware Resilience automatically alerts you and provides you with the information you need to evaluate the incident and respond quickly. By proactively identifying breaches, NetApp customers can block further unauthorized transfer of sensitive data, limiting the damage.

In this activity, you assume the role of a malicious insider and attempt to extract a large volume of data from a share protected by NetApp Ransomware Resilience. You will then resume your work as a storage administrator and, taking advantage of the User behavior analytics you enabled earlier in this lab, discover the breach in-progress and attempt to stop it.

1. From the Jumpshot taskbar, click the **File Explorer** icon.
2. In File Explorer, click **svm1_marketing** from the menu on the left to navigate to the share of svm1_marketing.

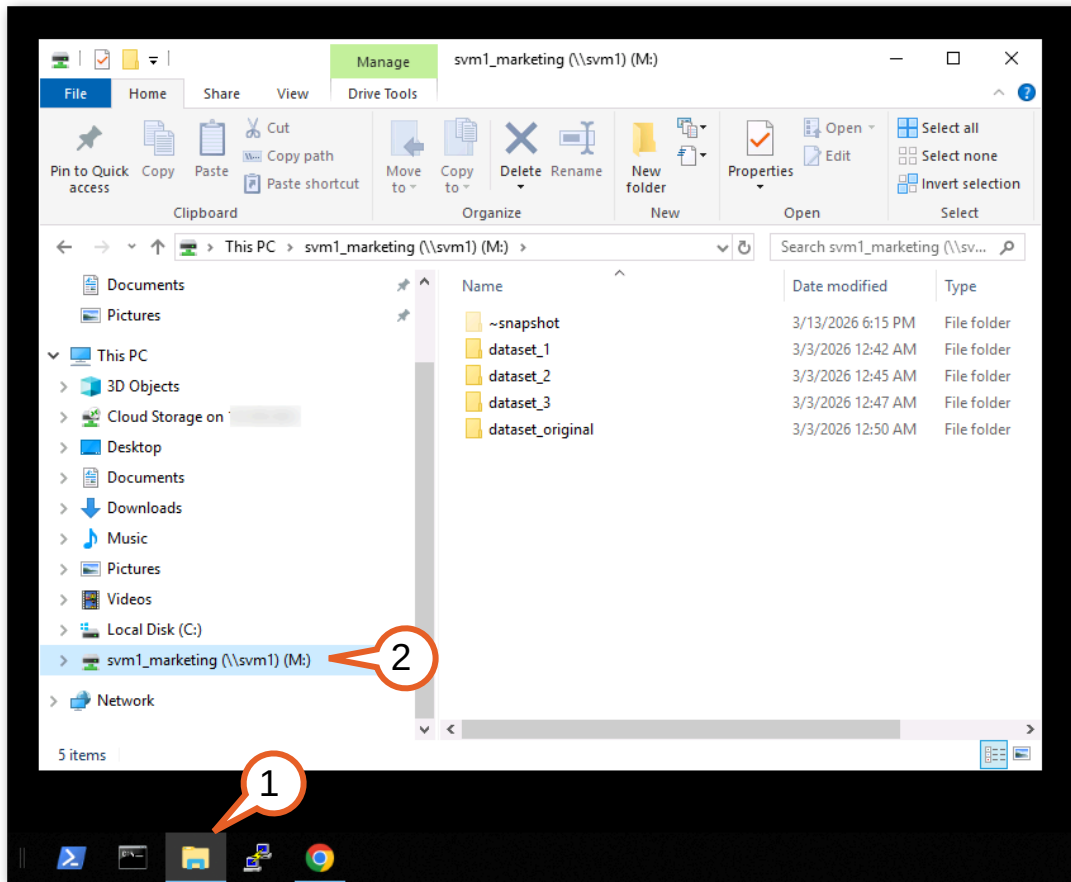


Figure 2-90:

3. Your goal is to extract data from the dataset_3 directory. Right-click **dataset_3** and then select **Copy** from the context menu.

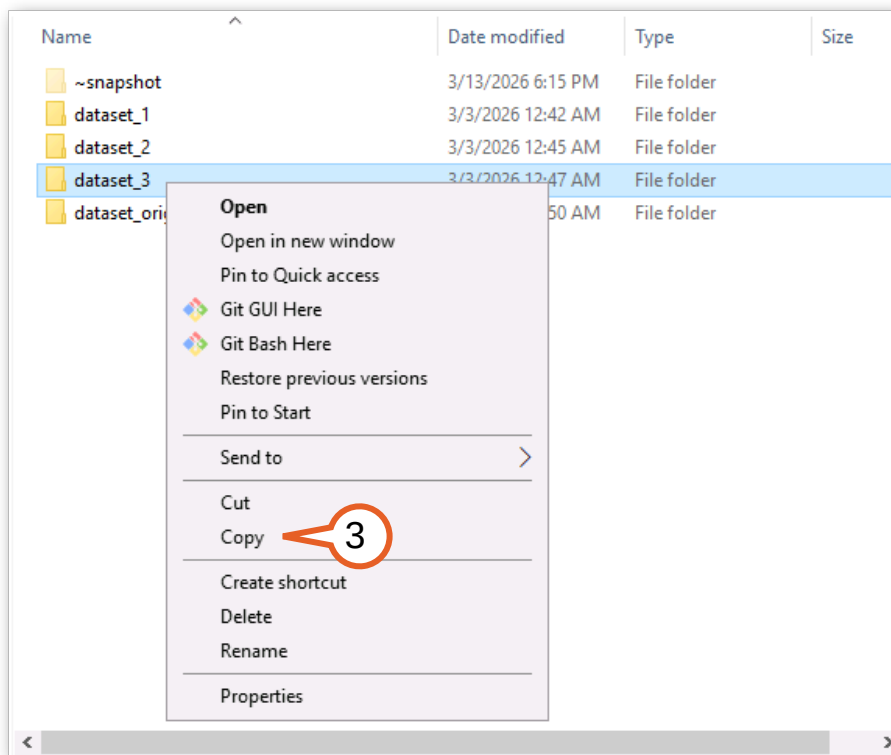


Figure 2-91:

4. Click **Documents** from the menu on the left.

5. Right-click in the white-space of the documents folder and select **Paste** from the context menu.

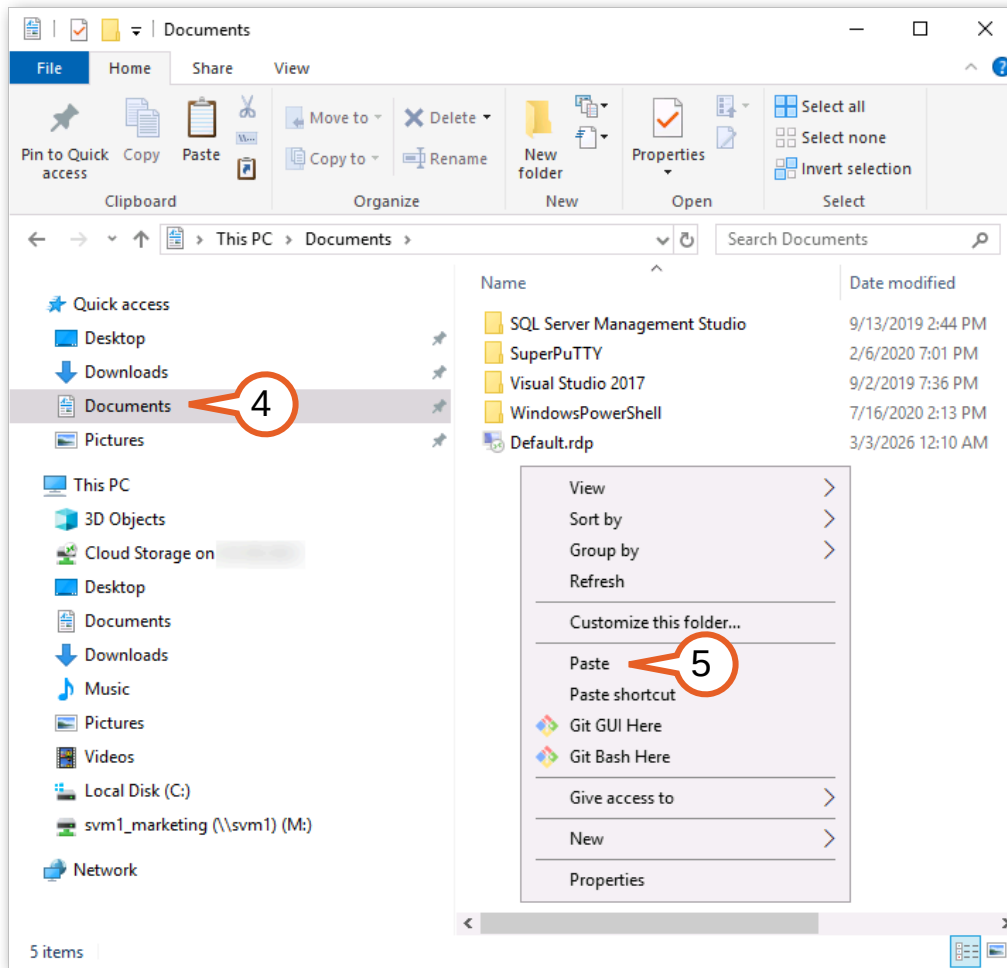


Figure 2-92:

6. A small window appears displaying the status of the copy job.

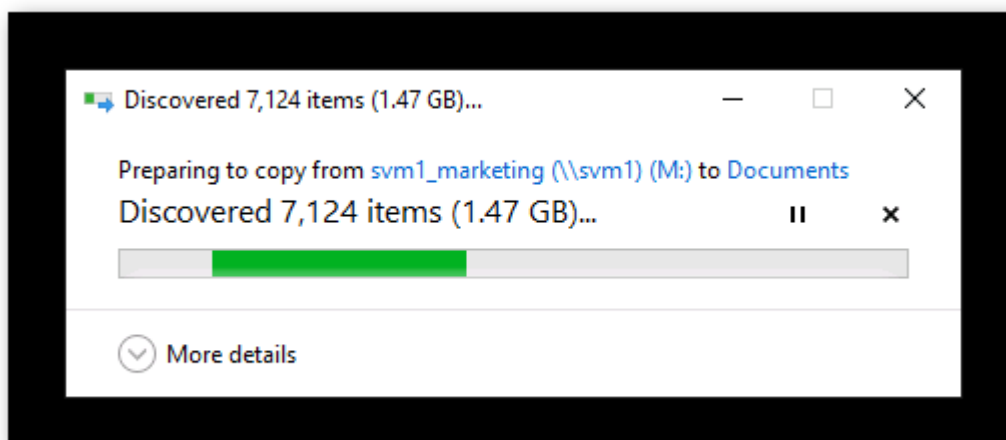


Figure 2-93:

7. With the exfiltration underway, leave the copy status window open and resume your role as storage administrator. Return to Ransomware Resilience, and click the **Alerts** tab.

Wait for an alert to appear in this view.



Important: In this scenario, it can take up to 15 minutes for Ransomware Resilience to detect the ransomware activity.

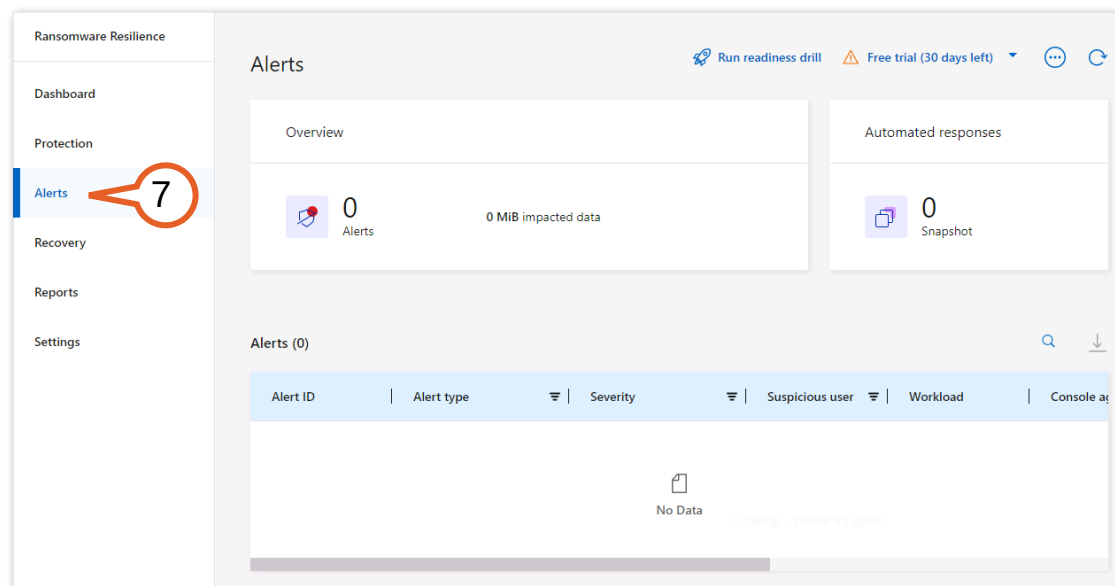


Figure 2-94:

8. When the alert appears, notice there is also a red badge on the NetApp Console notifications icon. Click the **notifications icon**.
9. The potential ransomware attack alert is surfaced here and can also be emailed to administrators of this NetApp Console organization (depending on your NetApp Console notification settings).

Click **outside the notification panel** to dismiss it.



Tip: NetApp Ransomware Resilience also lets you connect to a security information and event management (SIEM) system for threat reporting. You can configure this by clicking the menu icon at the top-right of the page and navigating to Settings, and then clicking **Connect** under SIEM connection.

Additionally, within ONTAP, when ARP detects abnormal volume activity, it generates an Event Management System (EMS) notification which can be forwarded directly to an email address, a syslog server, or a Simple Management Network Protocol (SNMP) trap host.

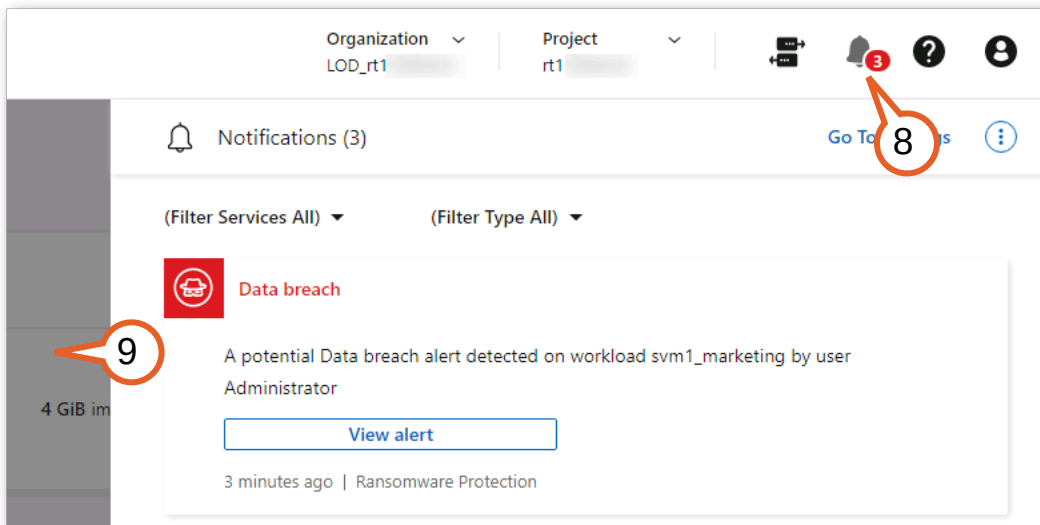


Figure 2-95:

10. In the alert table, there are multiple possible alert types including:

- Encryption
- Data Breach
- Data Destruction
- Suspicious user behavior

From the Alerts table, locate the alert with “Data Breach” as the Alert type and “svm1_marketing” as the Workload and click the **alert ID**.

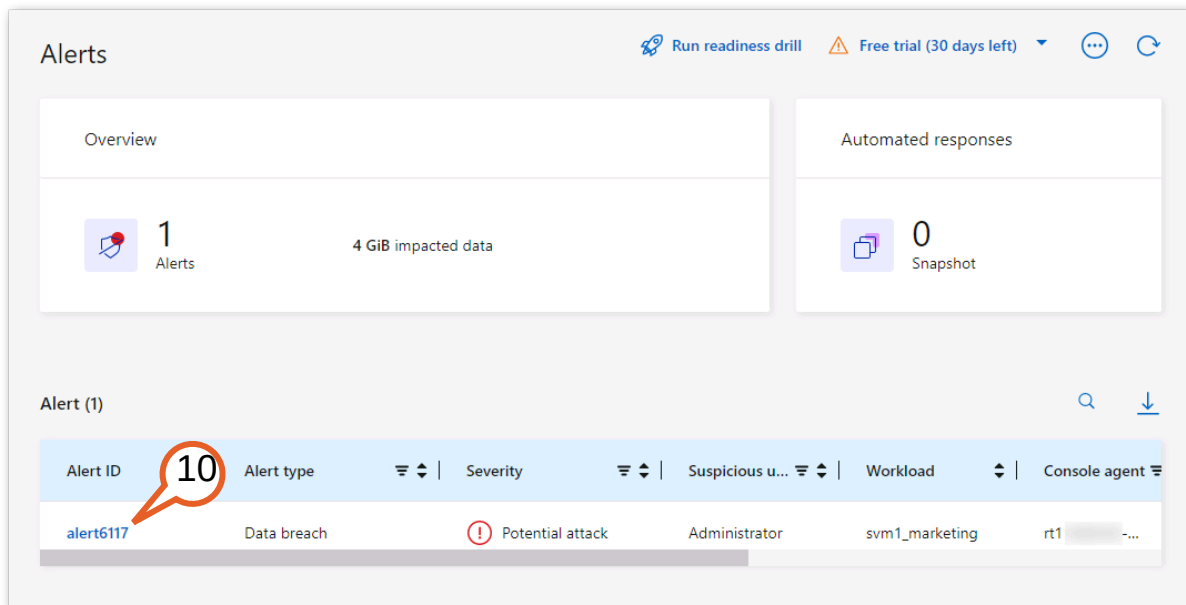


Figure 2-96:

11. The alert view provides information about where the attack occurred, when it was first detected, and how much data is impacted.

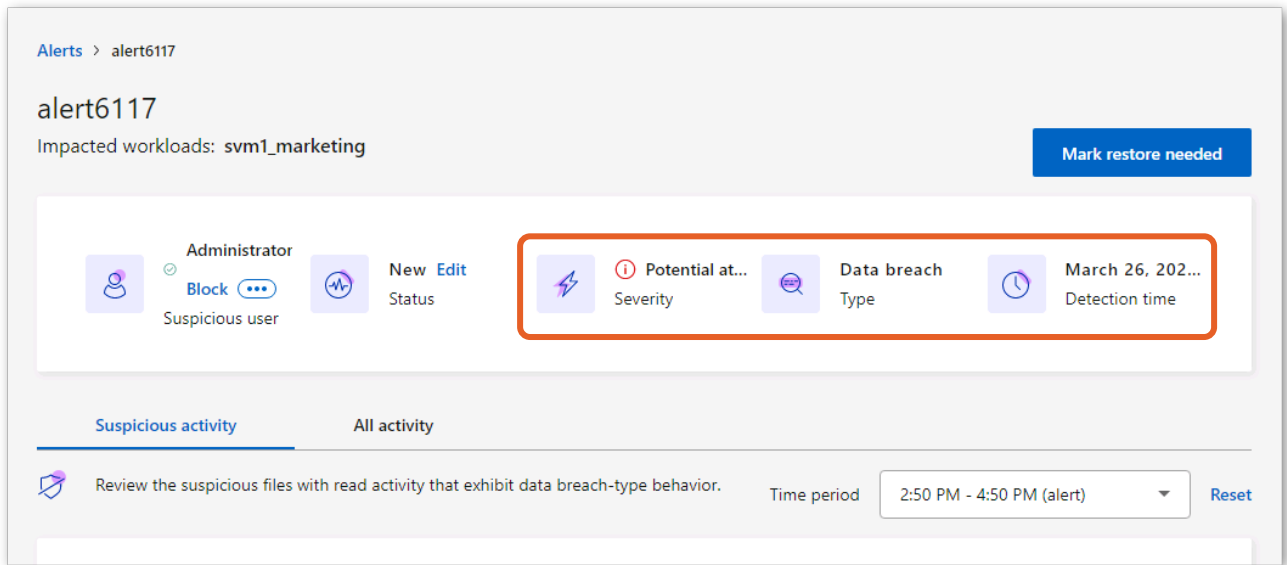


Figure 2-97:

12. The Suspicious activity tab is selected by default and displays a graph of read activity in the svm1_marketing volume over time, clearly showing an anomalous spike in activity.

Beneath that is information about impacted workloads, volumes, directories, and files, providing more detail about what data is being exfiltrated.

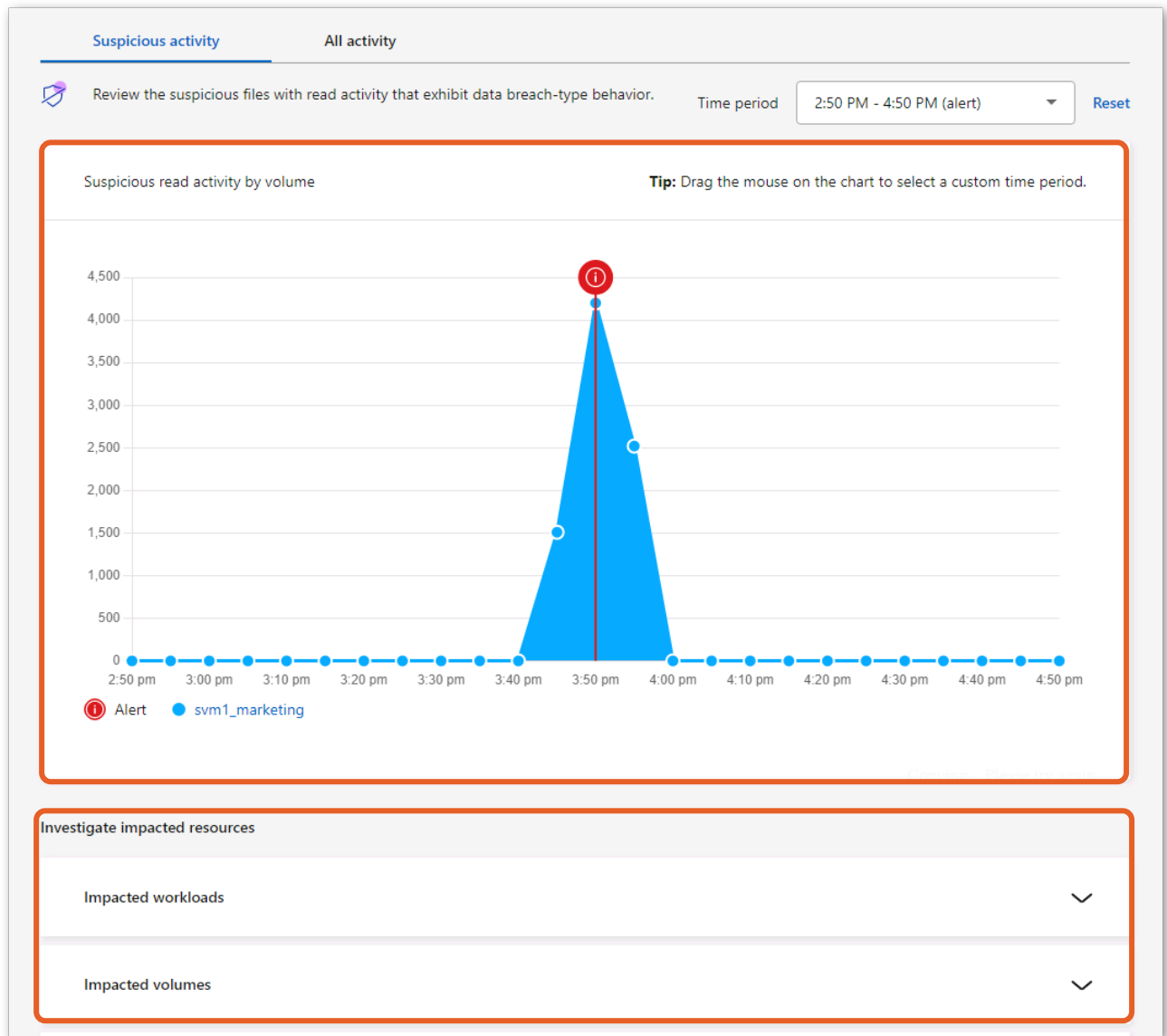


Figure 2-98:

13. Scroll back up and click **All activity**.
14. This tab shows you additional file activity over time, allowing you to visually assess more about the nature of the incident. For instance, if the increased read activity is paired with deletes or writes it could be a data destruction attack or a ransomware attack.

15. With sufficient evidence observed, you want to stop this activity as soon as possible.

At the top you may see some information about any identified suspicious users associated with the attack.

Click **Block** to block this users access to the file system.

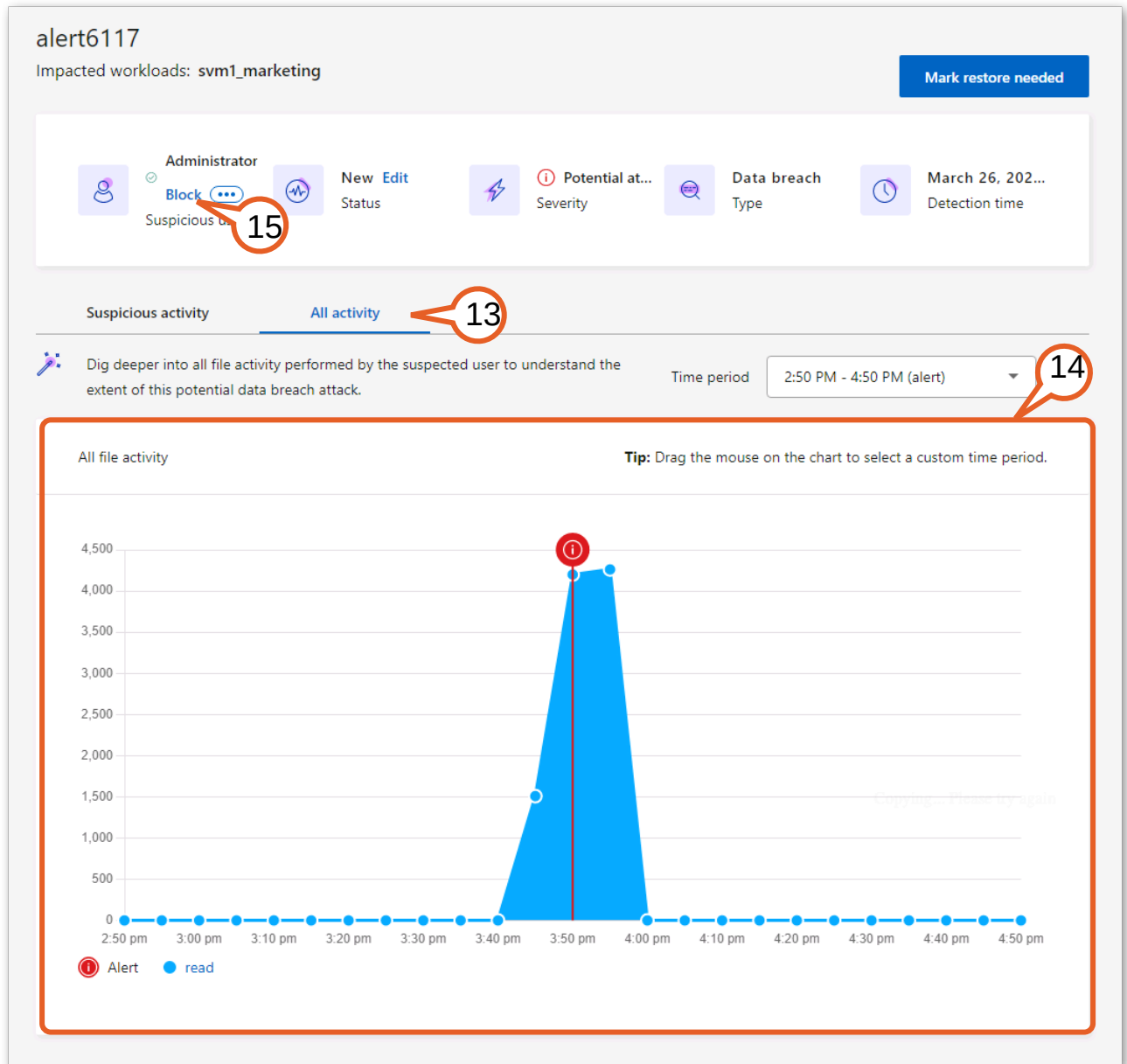


Figure 2-99:

16. In the confirmation pop-up, click **Block**.

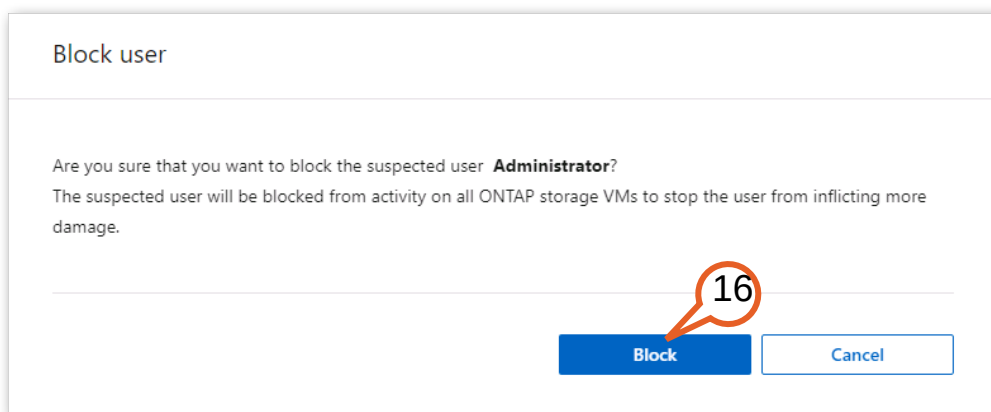


Figure 2-100:

17. Return to copy window. If your copy is still in-process, within a minute you should see a pop-up indicating the copy was interrupted. Your access to the filesystem has been revoked.

Click **Cancel** in the pop-up.

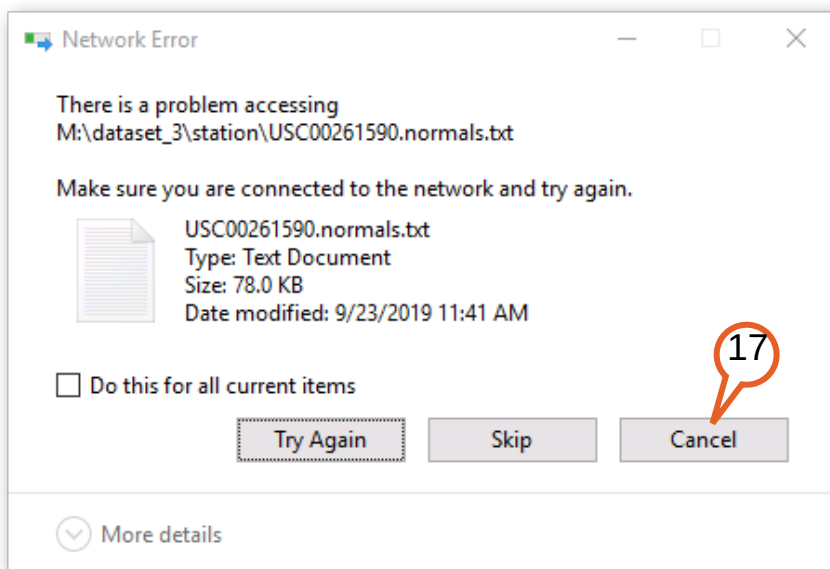


Figure 2-101:

Takeaway: NetApp Ransomware Resilience helps you rapidly respond to a suspected data breach by providing you with timely alerts, all the information you need to assess the situation, and an effective way to quickly stop the breach before more data is extracted.



Tip: In cases where the data breach involved data destruction or modification, you would click the **Mark restore needed** button in the top-right of the alert view and then proceed with a restore workflow, restoring any deleted or modified data using the restore points created with your protection strategy. To try out this workflow, take this lab again and choose [Option 1: Respond to a Ransomware Attack](#).

3 Lab Summary

In this lab, you used NetApp Ransomware Resilience to quickly discover, protect, and recover your workloads across many environments. After a simple setup process, you can easily take advantage of multiple NetApp ransomware protection technologies across many workloads in a fast and convenient interface.



Tip:

To see more NetApp cyber resilience solutions, check out these labs:

- “Ransomware Protection for NAS Workloads”
- “Ransomware Protection for SAN Workloads”
- “Ensure Data Retention with SnapLock and Snapshot Locking”
- “Layered Ransomware Protection with a Cyber Vault Using SnapLock Compliance”

4 Additional Information

4.1 References

The following references were used to write this lab guide.

- [NetApp Ransomware Resilience documentation](#)
- [User activity detection in NetApp Ransomware Resilience](#)
- [Autonomous Ransomware Protection documentation](#)
- [Ransomware Protection and Ransomware Recovery Guarantee](#)
- [Learn about ONTAP FPolicy solutions](#)
- [DataLock documentation](#)
- [Data Infrastructure Insights](#)
- [Workload Security](#)

4.2 Version History

Version	Date	Document Version History
1.0	June 2024	Initial Release.
1.1	September 2024	Updated to Ubuntu BlueXP connector, minor improvements
1.2	November 2024	Updated for BlueXP IAM and fixes from 3.9.46 standard mode release
1.3	January 2025	Updated discovery process
1.4	April 2025	Updated to ONTAP 9.16.1, enhanced activities
2.0	October 2025	Rebranded to Ransomware Resilience and NeApp Console in Demo Mode
2.0.1	December 2025	Corrected demo workload.
3.0	March 2026	Switch from demo mode to real lab resources
3.0.1	April 2026	Minor improvements

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